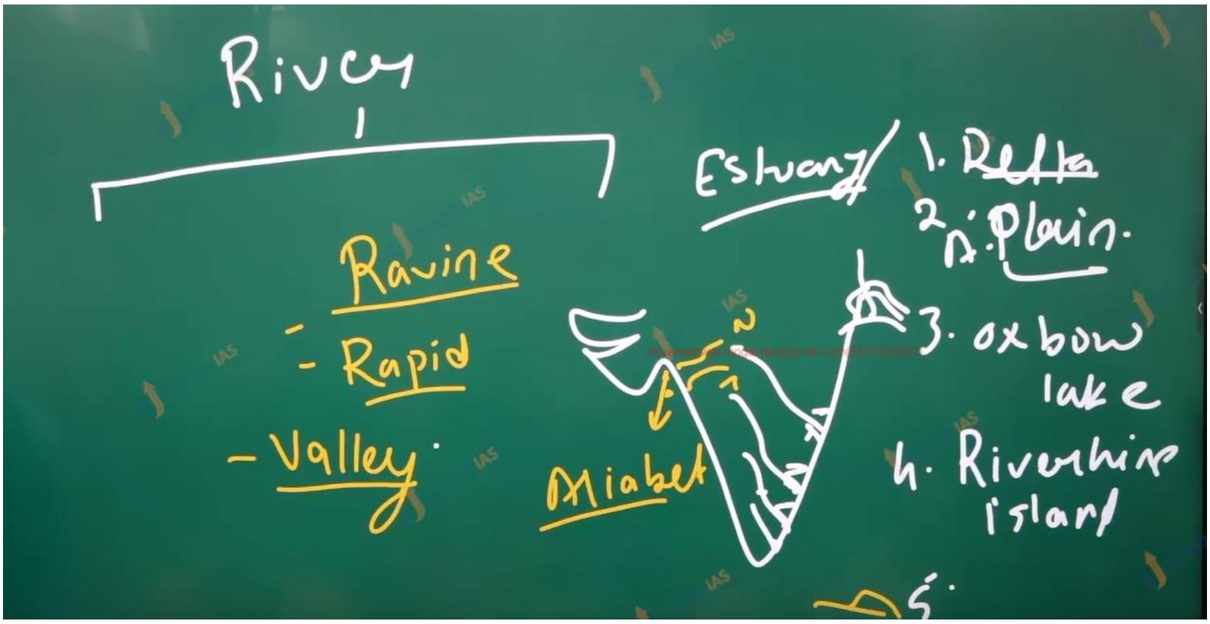
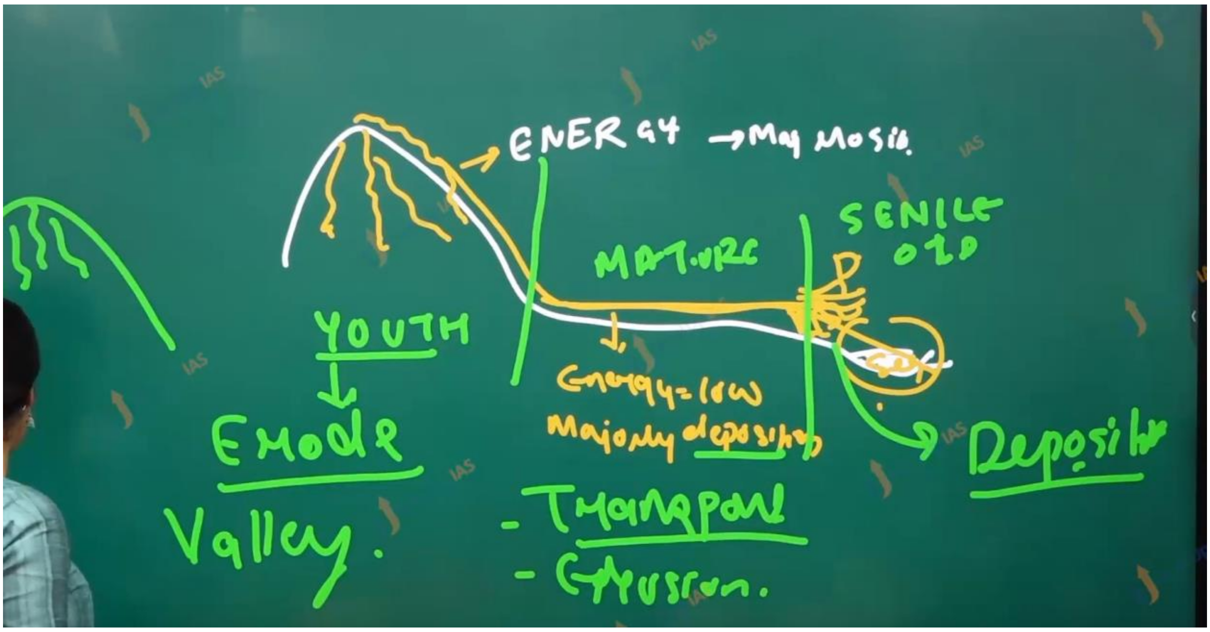
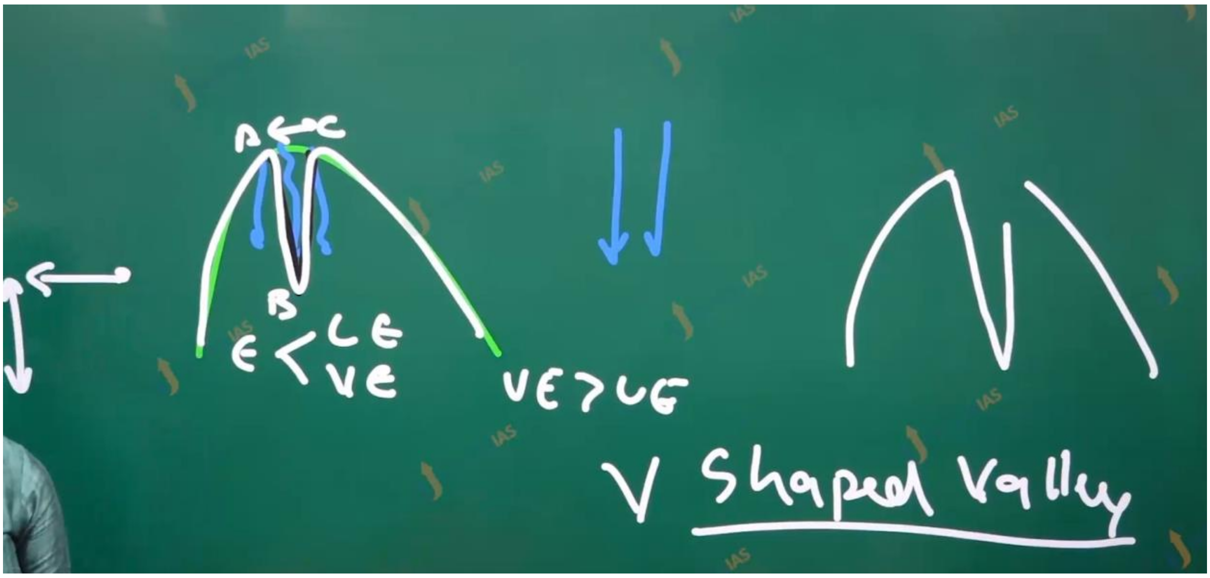
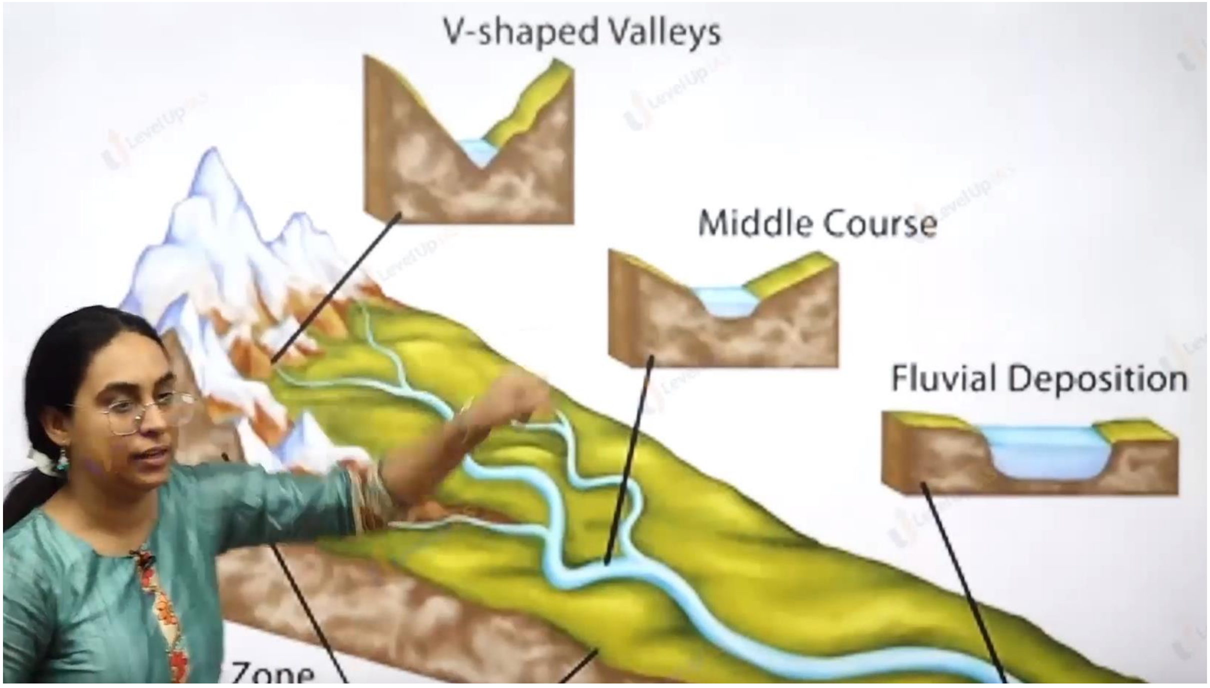
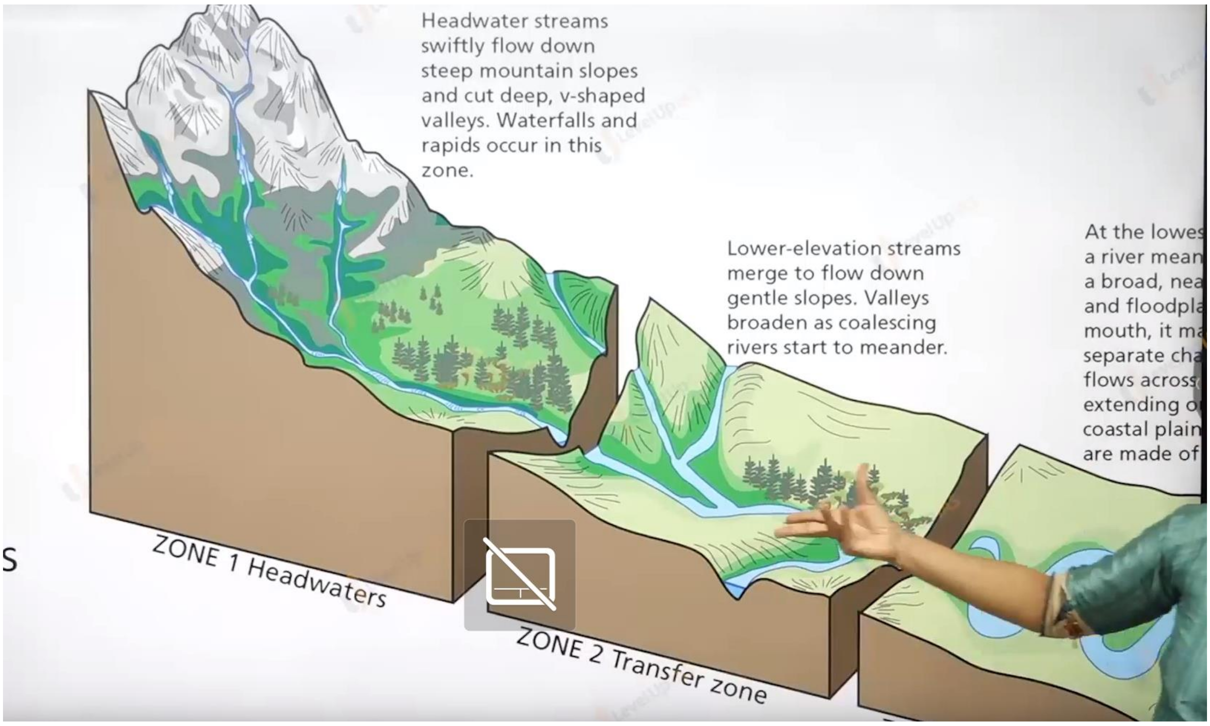
Landforms

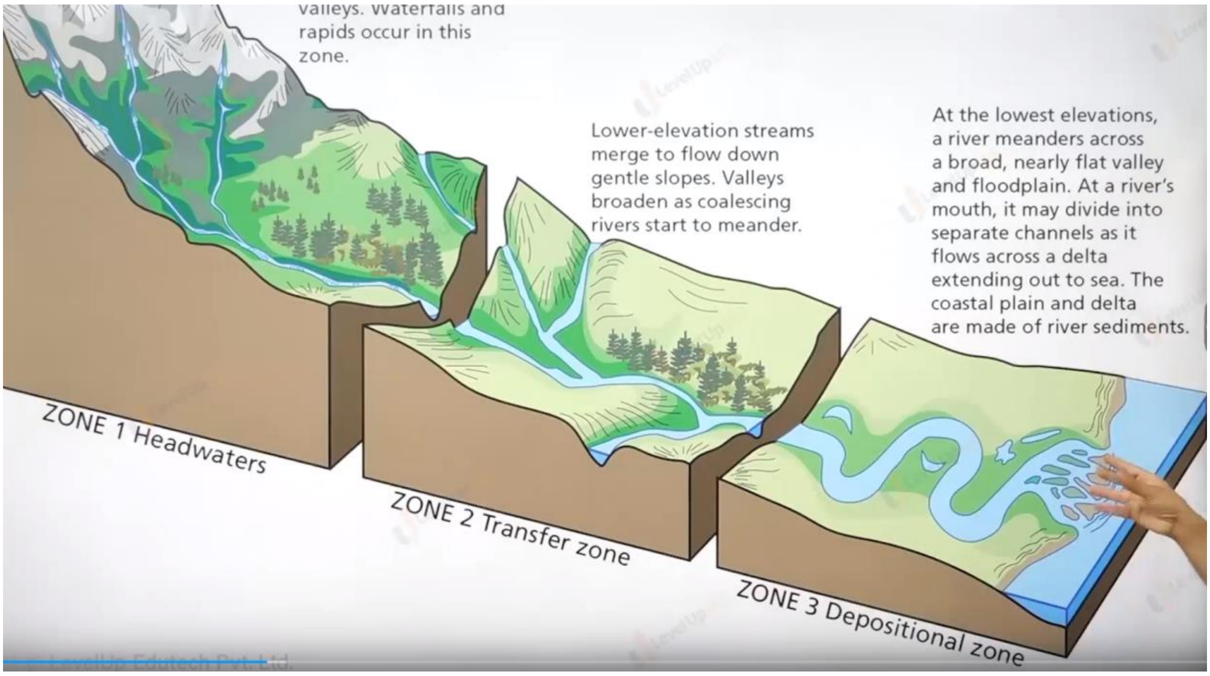

River
Estuary
1. Delta
3. ox bow lake
4. Riverine island
Ravine
- Rapid
- Valley.

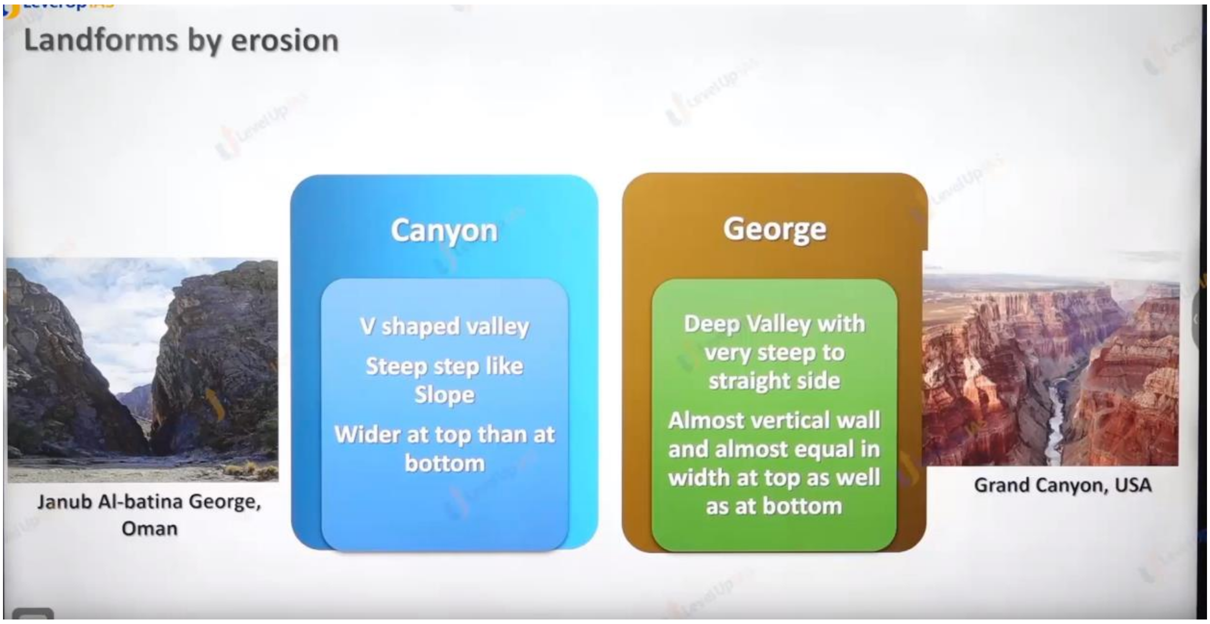

ENERGY
MATURE
SENILE
OLD
YOUTH
Erode
Valley.
Energy = low
- Transport


A
C
B
LE
E
VE
VE>VG
V Shaped Valley


V-shaped Valleys
Middle Course
Fluvial Deposition
Zone


Headwater streams swiftly flow down steep mountain slopes and cut deep, v-shaped valleys. Waterfalls and rapids occur in this zone.
Lower-elevation streams merge to flow down gentle slopes. Valleys broaden as coalescing rivers start to meander.
At the lowes
a river mean
a broad, nea
and floodpla
mouth, it ma
separate cha
flows across
extending o
coastal plain
are made of
S
ZONE 1 Headwaters
ZONE 2 Transfer zone

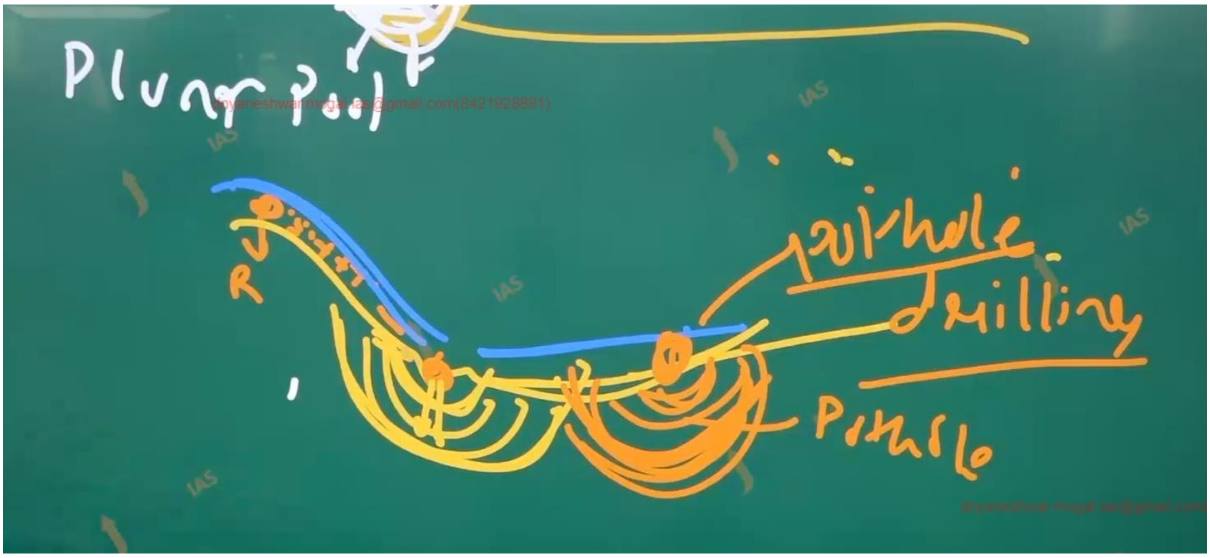

valleys. Waterfalls and rapids occur in this zone.
Lower-elevation streams merge to flow down gentle slopes. Valleys broaden as coalescing rivers start to meander.
At the lowest elevations, a river meanders across a broad, nearly flat valley and floodplain. At a river's mouth, it may divide into separate channels as it flows across a delta extending out to sea. The coastal plain and delta are made of river sediments.
ZONE 1 Headwaters
ZONE 2 Transfer zone
ZONE 3 Depositional zone

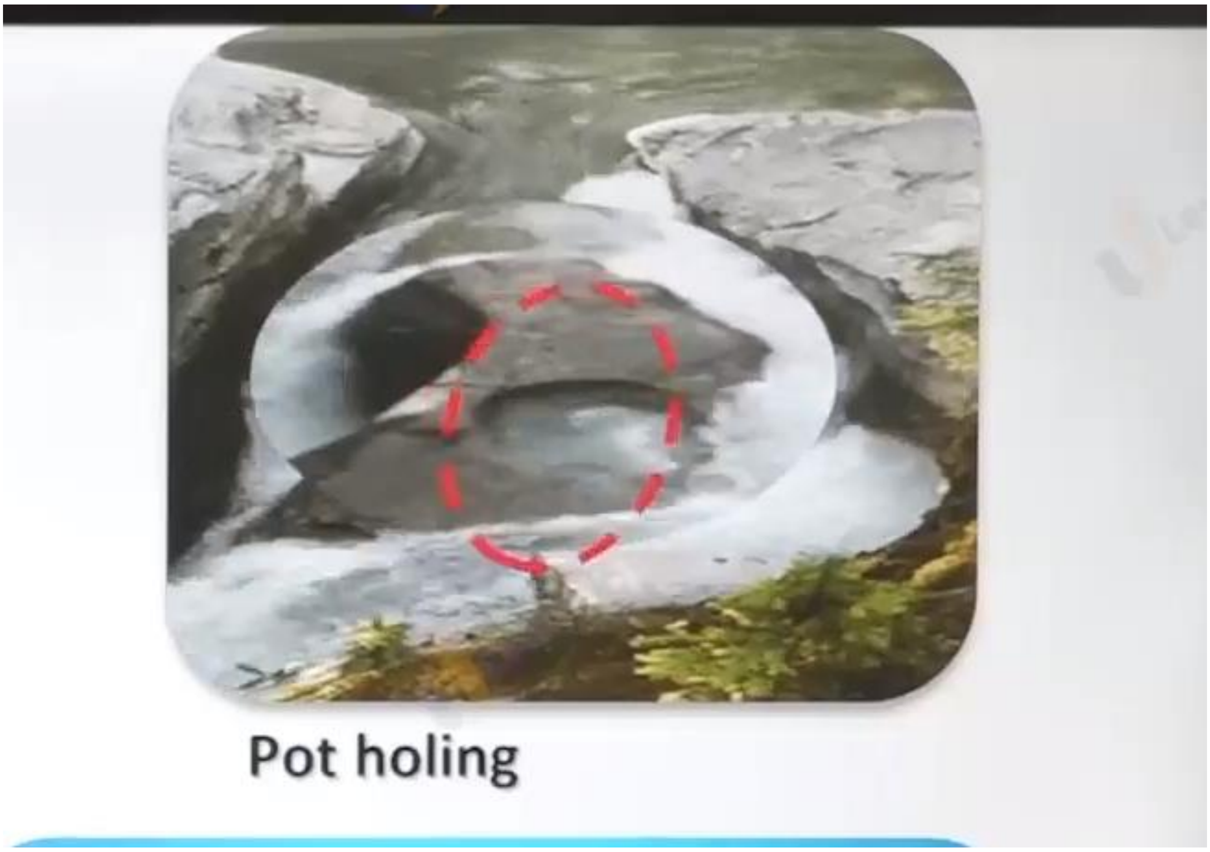


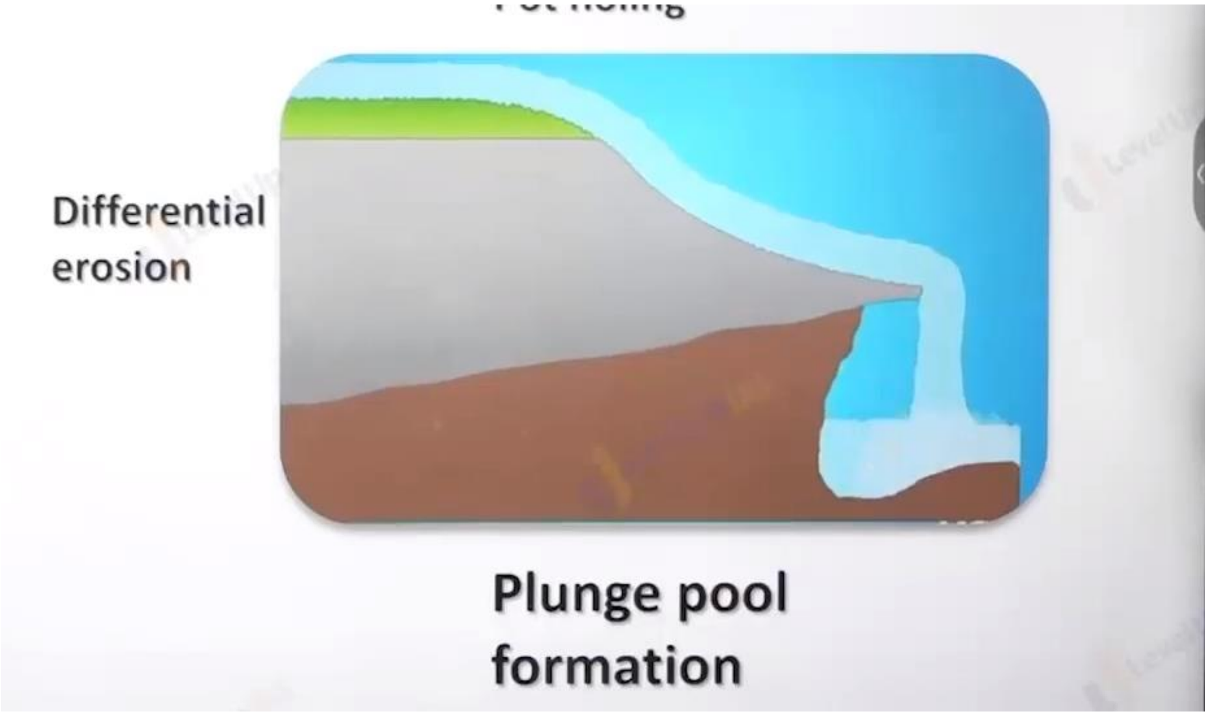
Erosion land from by the river:

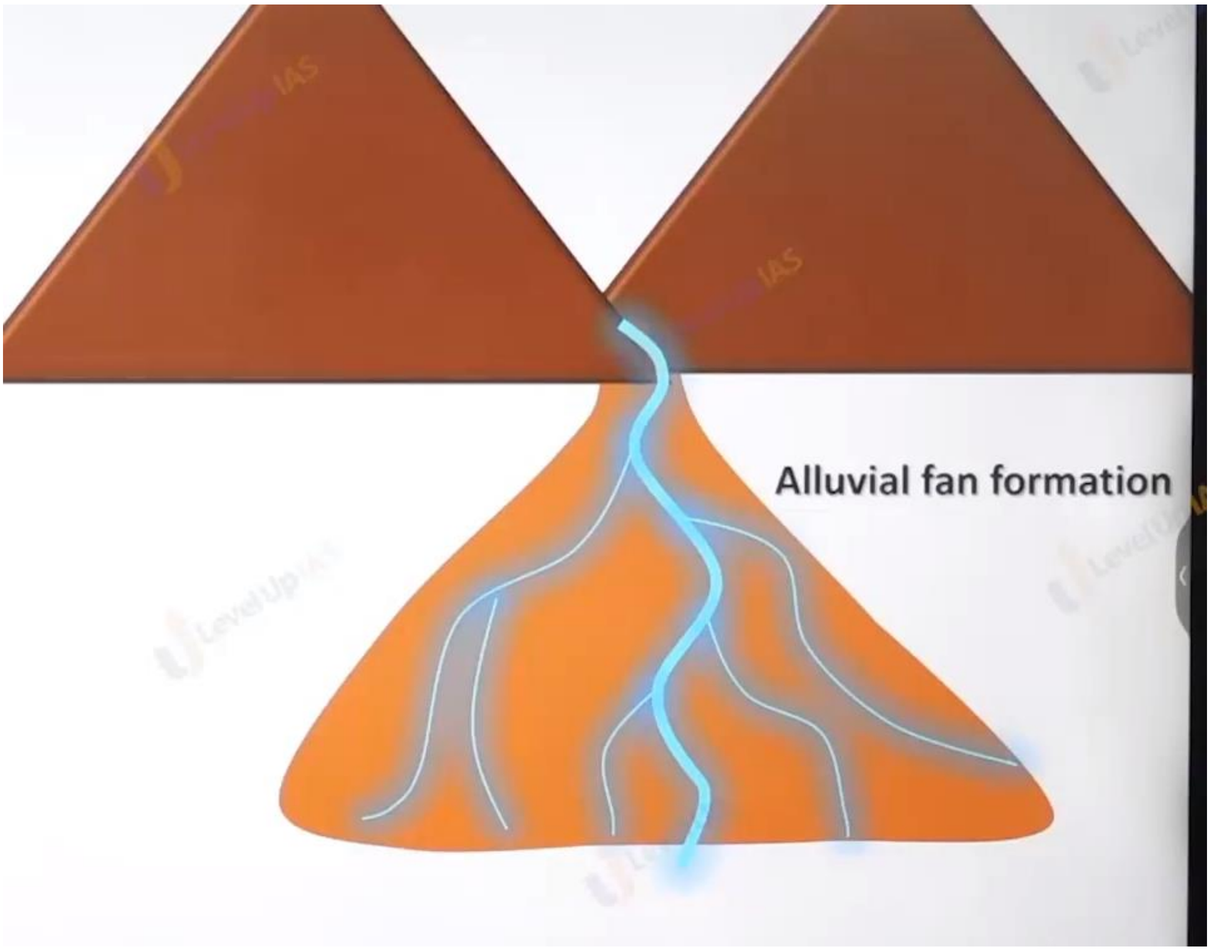
1. George
2. Canyon
3. Plunge pool and waterfall.
4. Pothole.

Pothole erosion and pothole drilling.



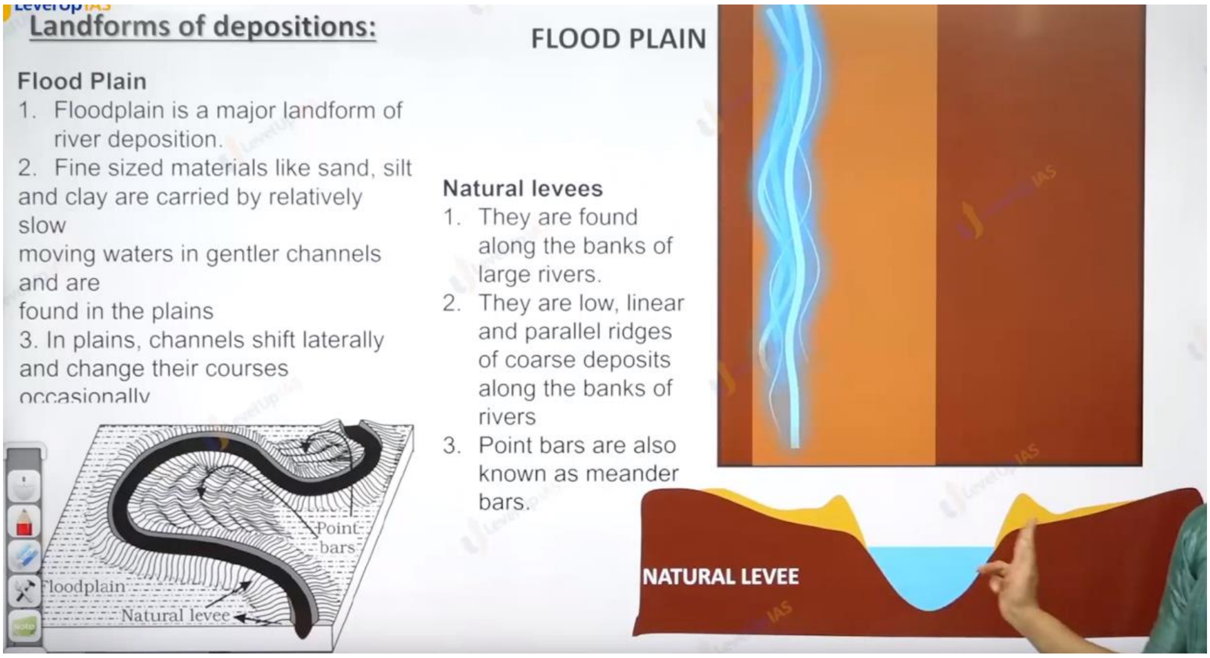


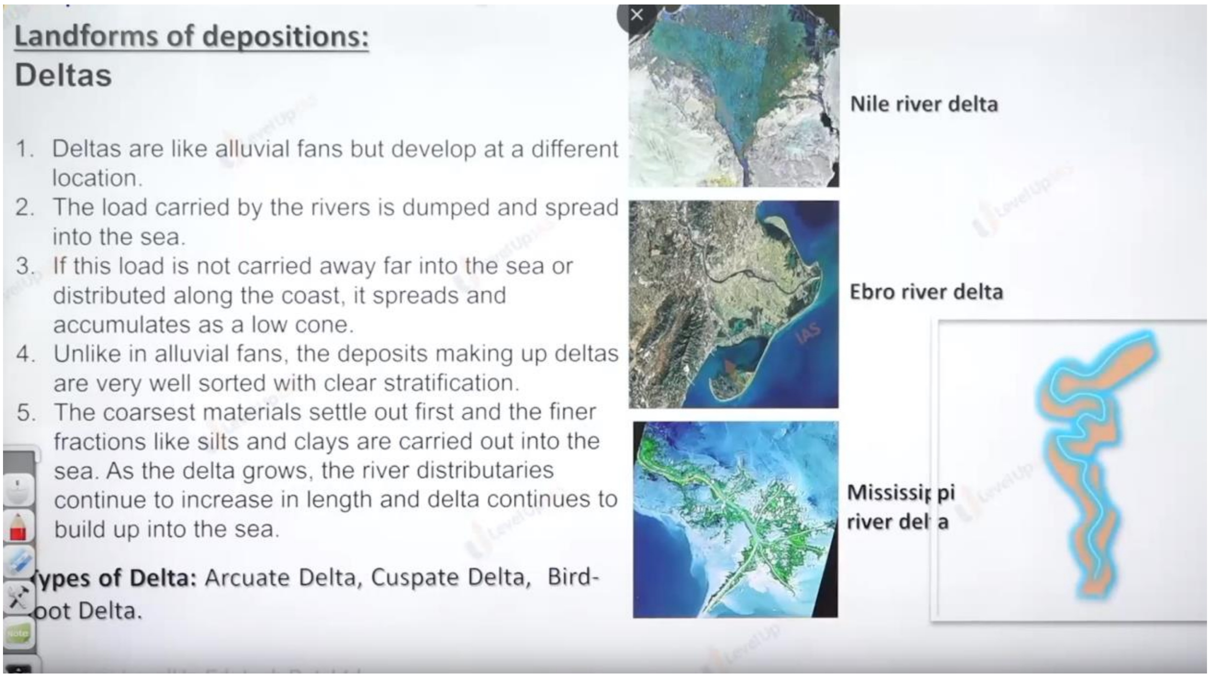


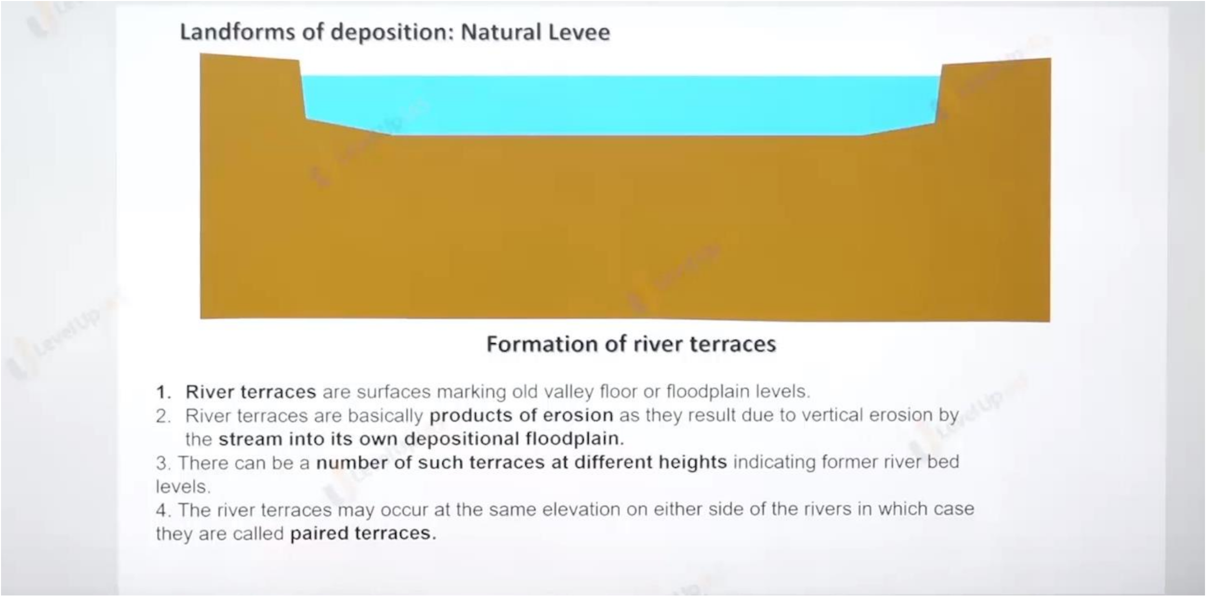


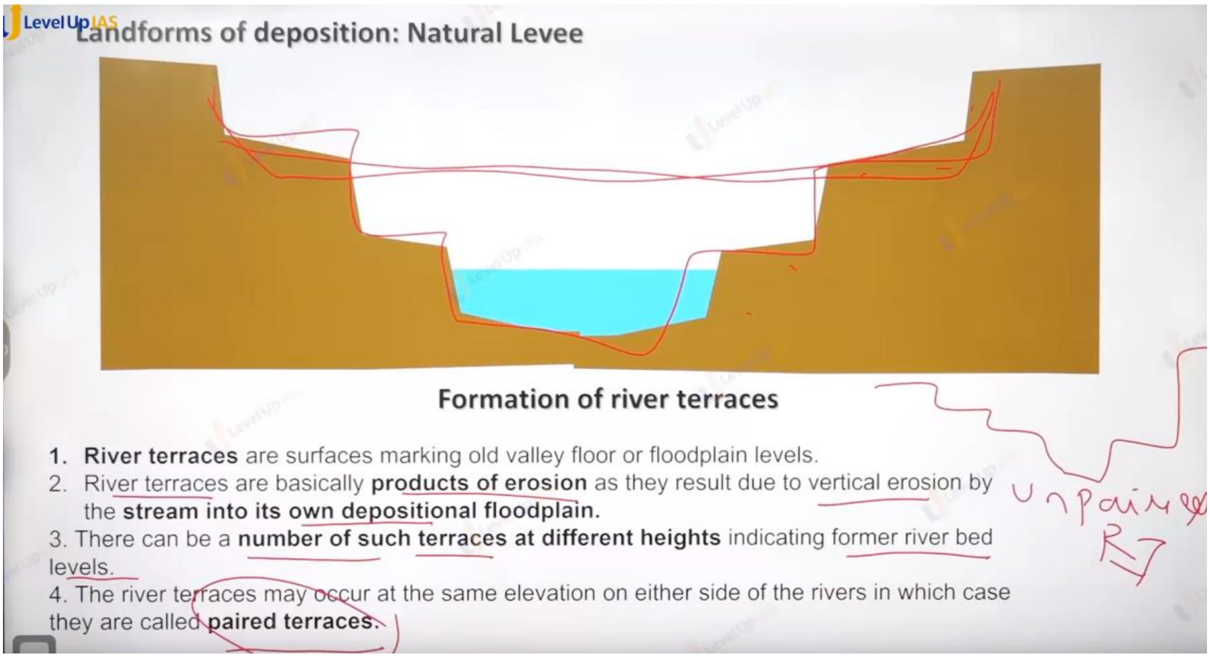
Deposition landform

1. Alluvial plain
2. Flood plain
3. Natural leavy
4. delta.

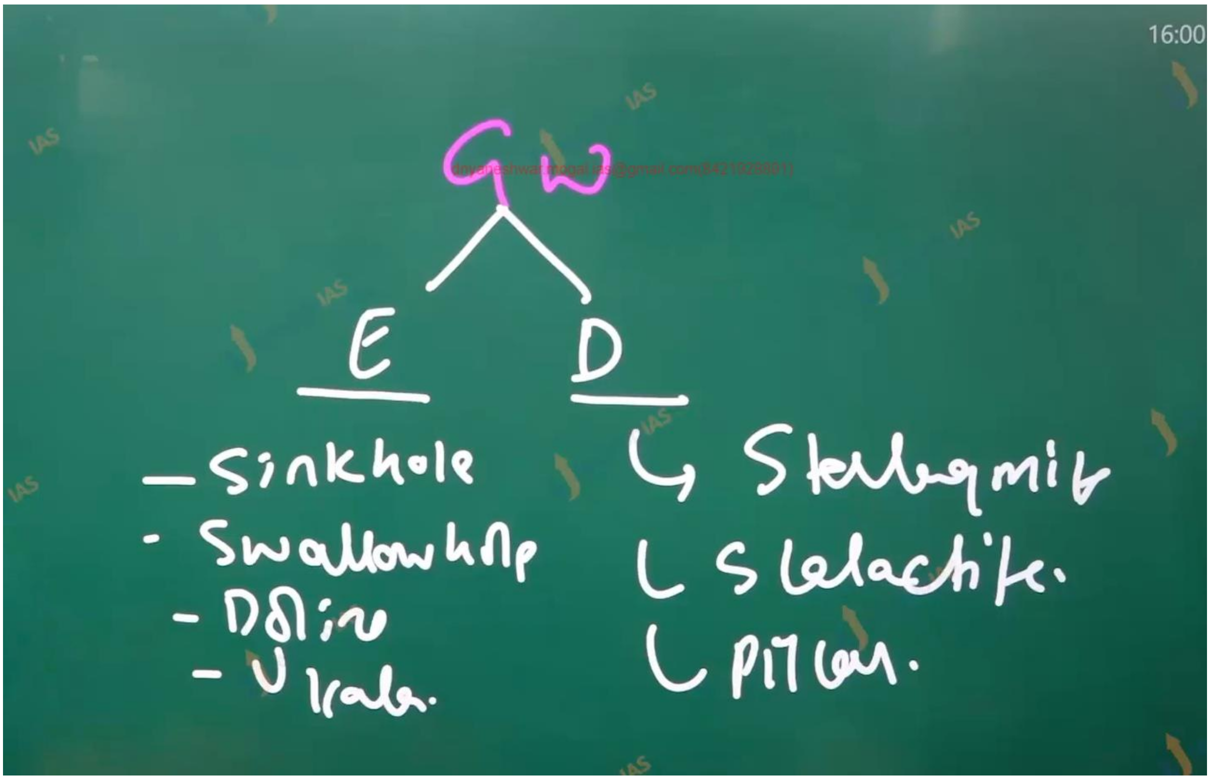




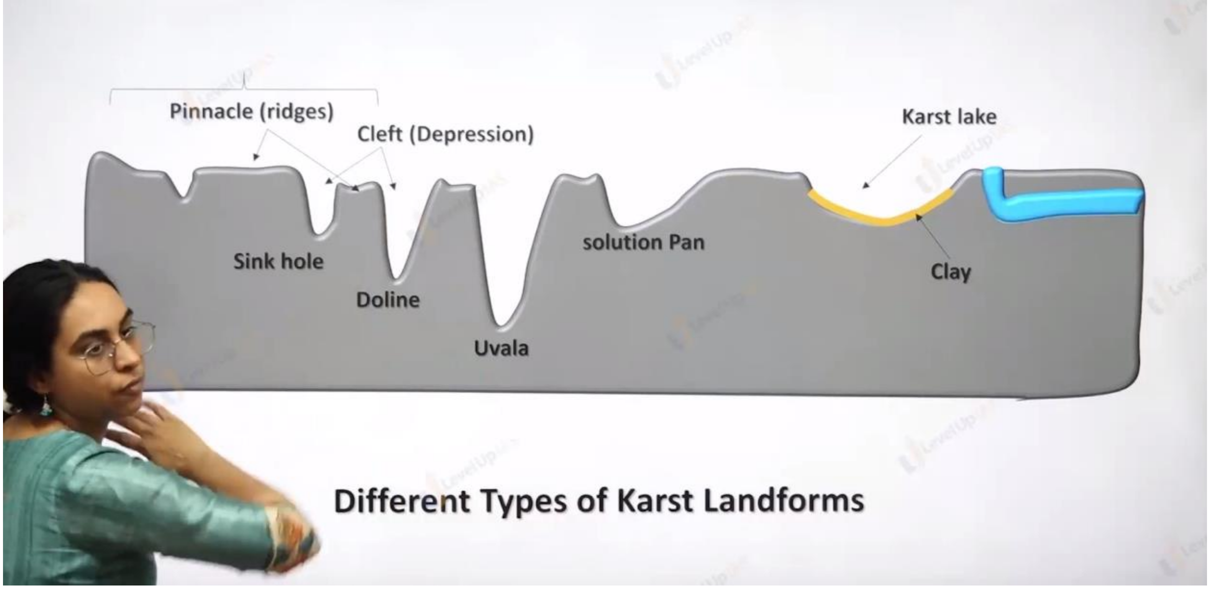


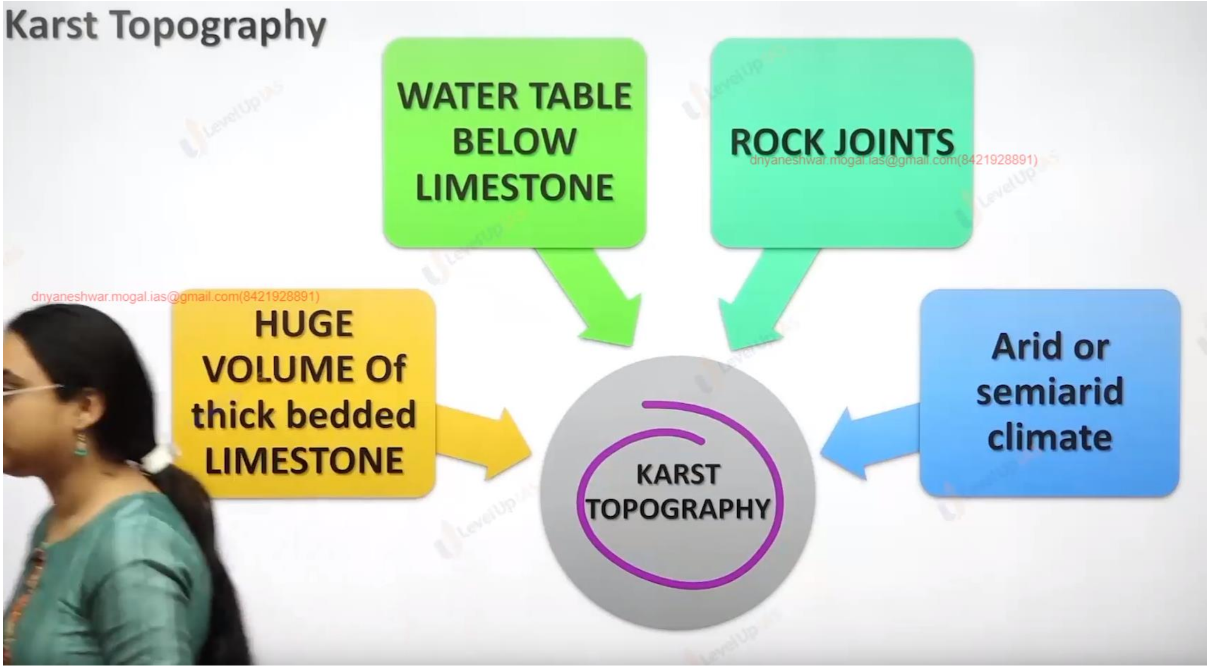


Ground water erosion.



Not all time erosion happened when it limewater erosion from.





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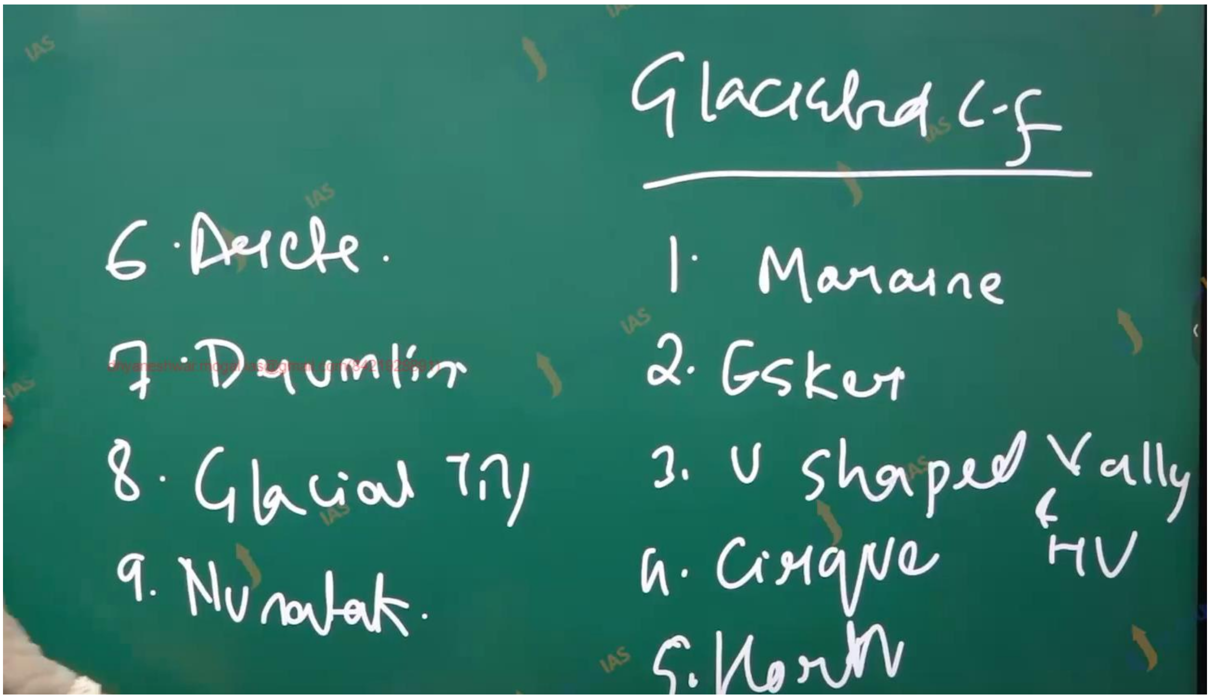
1. Cast erosional landforms:
2. sink hole
3. swallow hole.
4. Collapsing.
5. Uvala
6. Doline.
7. Limestone
8. Cave
9. Solution pan

10. Depositional landforms.

---

11. Stalactite hangs cave curtains.
12. Stalagmites rise up



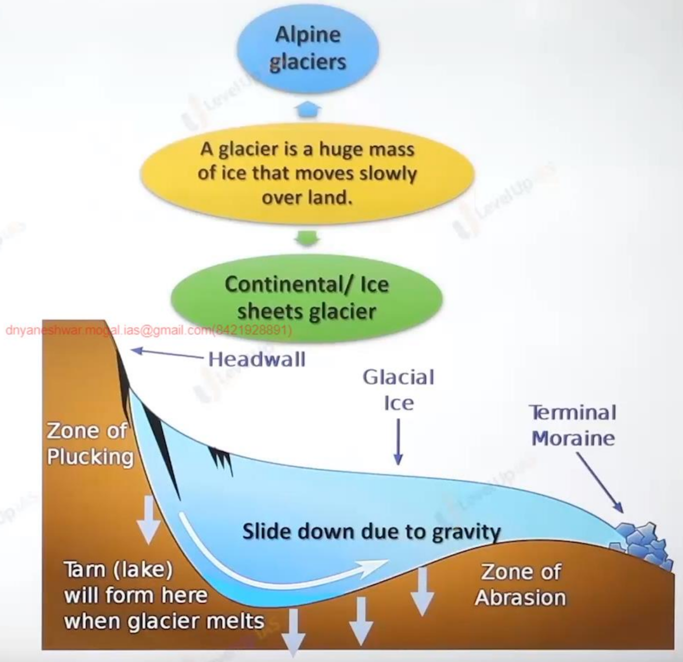
Masses of ice moving as sheets over the land (continental glacier or piedmont glacier if a vast sheet of ice is spread over the plains at the foot of mountains) or as linear flows down the slopes of mountains in broad trough-like valleys (mountain and valley glaciers) are called glaciers

The movement of glaciers is slow unlike water flow.

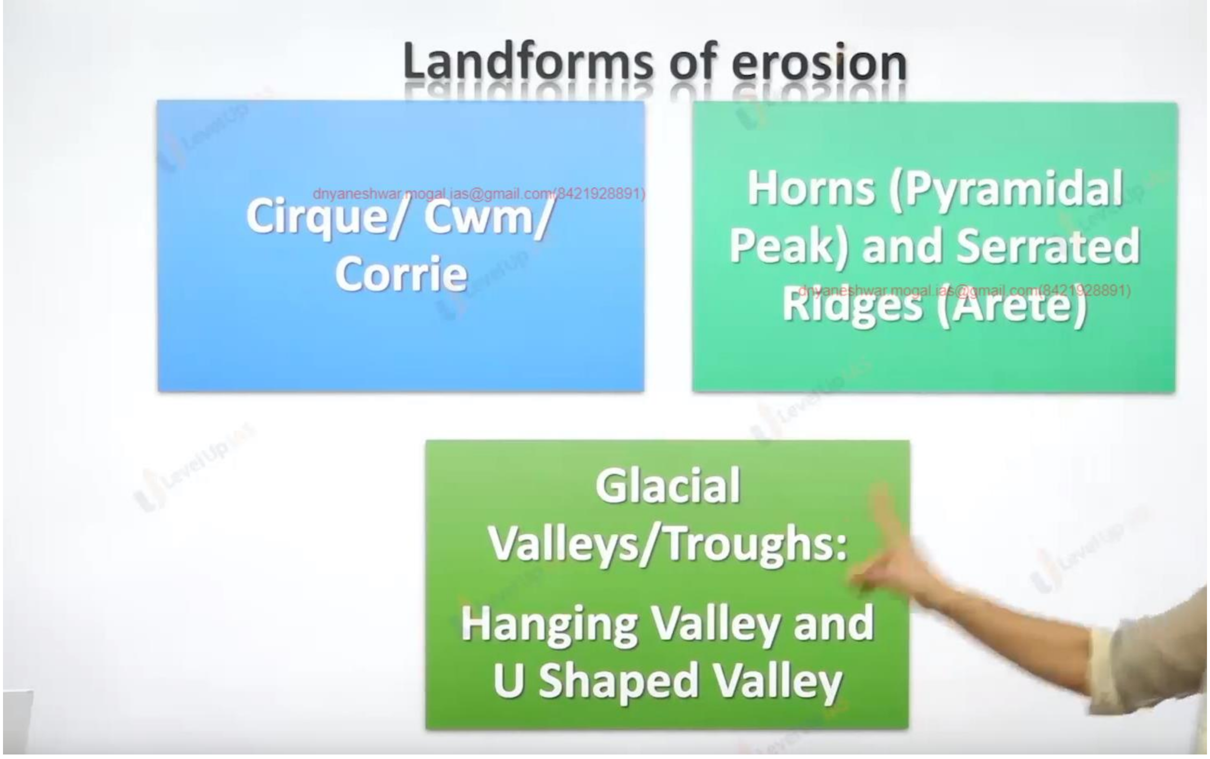
The movement could be a few centimetres to a few metres a day or even less or more.

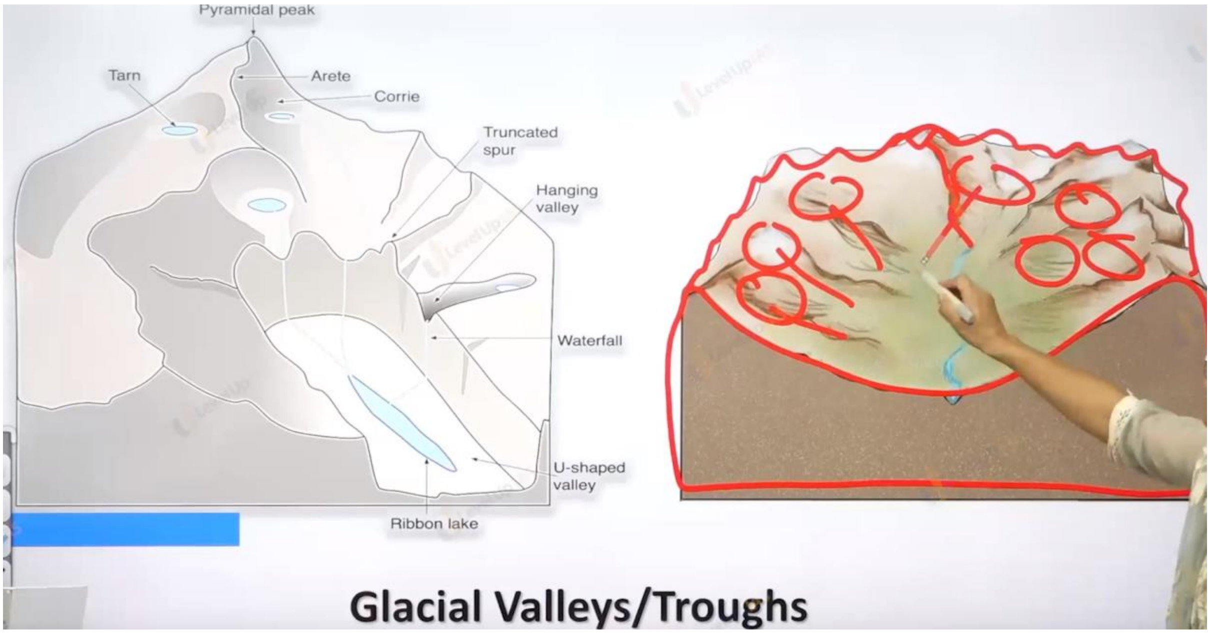
Glaciers move basically because of the force of gravity.

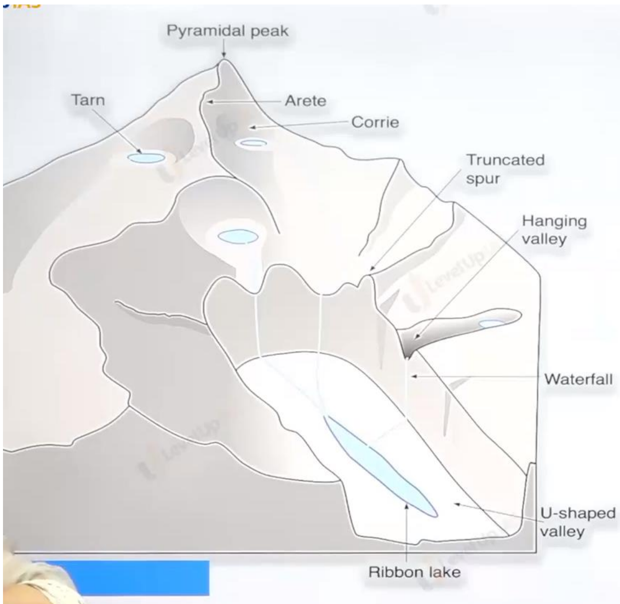
Erosion by glaciers is tremendous because of friction caused by sheer weight of the ice.



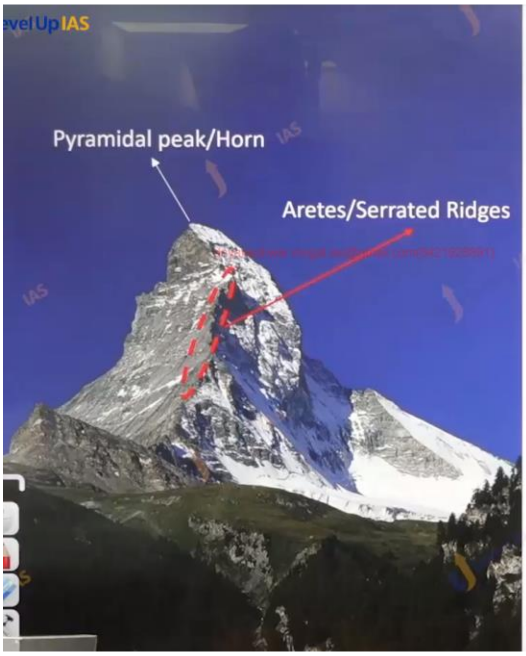
1. The material plucked from the land by glaciers (usually large-sized angular blocks and fragments) get dragged along the floors or sides of the valleys and cause great damage through abrasion and plucking.
2. Glaciers can cause significant damage to even un-weathered rocks and can reduce high mountains into low hills and plains.
3. As glaciers continue to move, debris gets removed, divides get lowered and eventually the slope is reduced to such an extent that glaciers will stop moving leaving only a mass of low hills and vast outwash plains along with other depositional features.


Landforms of erosion
Cirque/ Cwm/ Corrie
Horns (Pyramidal Peak) and Serrated Ridges (Arete)
Glacial Valleys/Troughs:
Hanging Valley and U Shaped Valley


Pyramidal peak
Tarn
Arete
Corrie
Truncated spur
Hanging valley
Waterfall
U-shaped valley
Ribbon lake
Glacial Valleys/Troughs



Glacial Valleys/Troughs



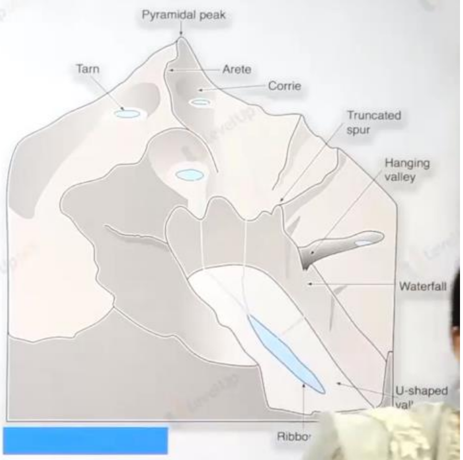
Pyramidal peak/Horn, Arete / Serrated Ridge

1. **Arete:** When two Corries cut back on the opposite side of the mountain a Knife-edged ridge/ Zigzag Ridge are formed called Arete or Serrated Ridge
2. **Horn:** When 3/more cirques cut back together they form angular horn or pyramidal peak.

The Matterhorn, part of the Alps in Switzerland, is a

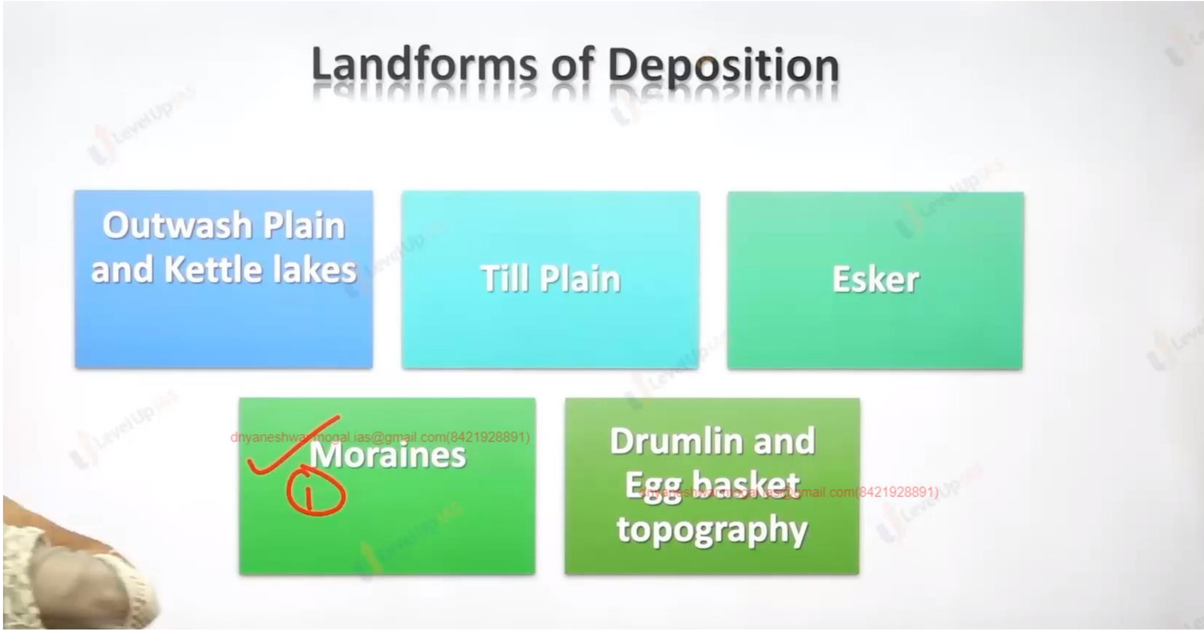
U shaped Valley

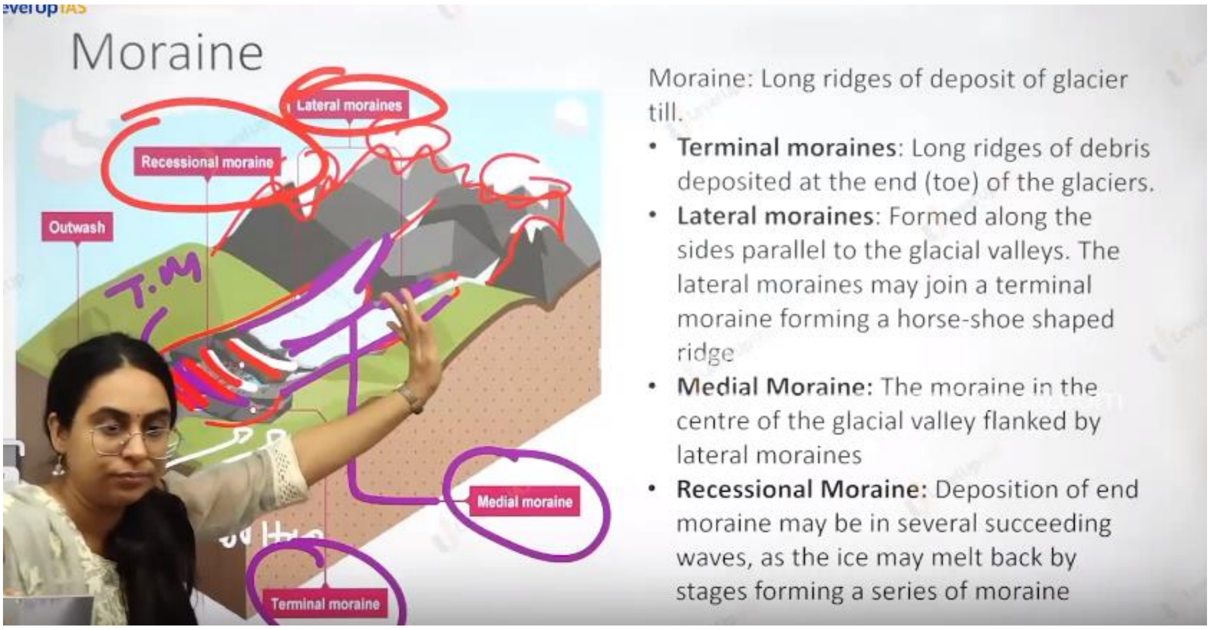
- Glacier on the downward journey fed by ice from several Corries like tributaries that join a river begins to wear away the sides and the floor of valley down
- It straightens any protruding spur on its course and the interlock spurs are blunted to form truncated spurs
- A U shaped valley is formed because of this process.
- After the disappearance of the ice, glacial valley may be filled with water forming Ribbon Lake

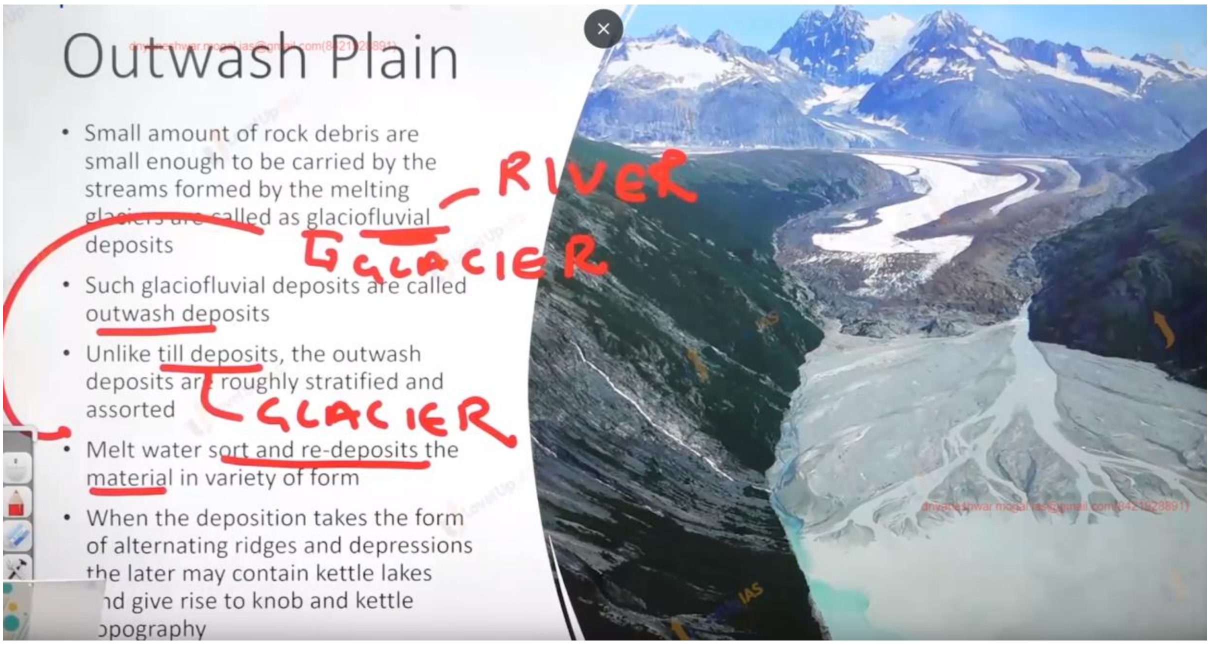


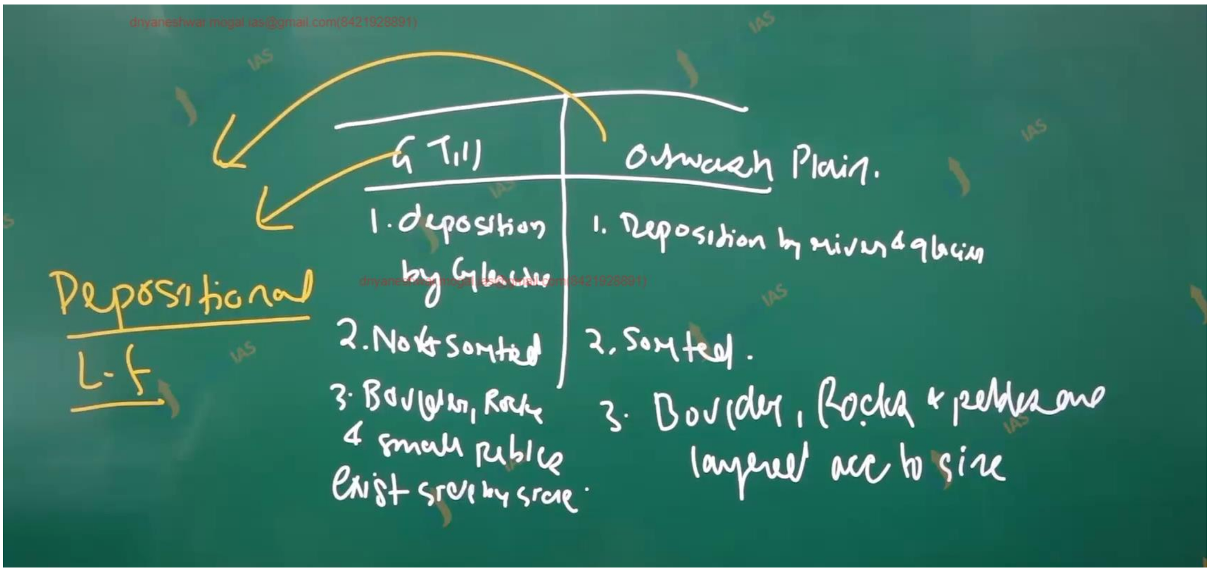
Hanging Valley

- The main valley is eroded much rapidly than the tributary valleys
- After the ice has melted, a tributary valley therefore hangs above the main valley.
- Search tributary valleys are termed as hanging valley


Landforms of Deposition
Outwash Plain and Kettle lakes
Till Plain
Esker
Moraines
1
Drumlin and Egg basket topography


Moraine
Lateral moraines
Recessional moraine
Outwash
T.M
Medial moraine
Terminal moraine
Moraine: Long ridges of deposit of glacier till.
• Terminal moraines: Long ridges of debris deposited at the end (toe) of the glaciers.
• Lateral moraines: Formed along the sides parallel to the glacial valleys. The lateral moraines may join a terminal moraine forming a horse-shoe shaped ridge
• Medial Moraine: The moraine in the centre of the glacial valley flanked by lateral moraines
• Recessional Moraine: Deposition of end moraine may be in several succeeding waves, as the ice may melt back by stages forming a series of moraine


Outwash Plain
• Small amount of rock debris are small enough to be carried by the streams formed by the melting glaciers are called as glaciofluvial deposits
RIVER
GLACIER
• Such glaciofluvial deposits are called outwash deposits
• Unlike till deposits, the outwash deposits are roughly stratified and assorted
GLACIER
• Melt water sort and re-deposits the material in variety of form
• When the deposition takes the form of alternating ridges and depressions the later may contain kettle lakes and give rise to knob and kettle topography


G Till
Outwash Plain.
1. deposition by Glacier
1. Deposition by river & glacier
Depositional L.F
2. Not sorted
2. Sorted.
3. Boulders, Rocks & small pebbles exist side by side.
3. Boulder, Rocks & pebbles are layered acc to size

Till Plain

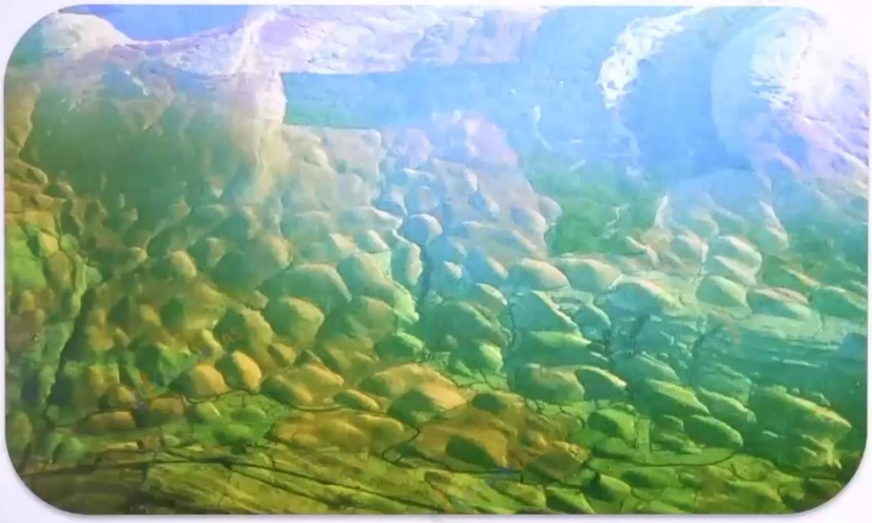
- The **unsorted** coarse and fine debris dropped by melting glacier is called as glacial till
- Most of the rock fragments in the angular to sub angular
- It is spread in sheets and not mounds
- Form Till Plain
- Landform is monotonous and featureless

Esker

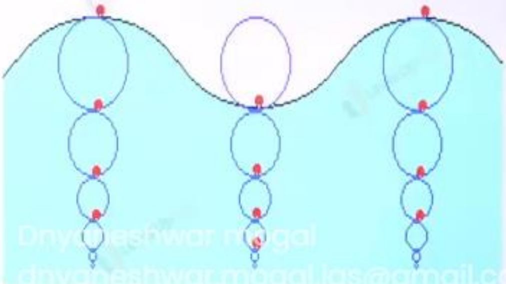
- They are long narrow sinous ridges composed of sand and gravel
- They mark the former site of subglacial meltwater streams
- They may be few feet in height but are several miles long
- they are made up of highly porous sand and gravel so water is rapidly drained off from their crest

Drumlin

- They are smooth Oval shaped features composed of glacial till along with gravel and sand
- The long axis of drumlin are parallel to the direction of the ice movement so drumlin indicate the direction of glacier movement
- One end of the drumlin facing the glacier are called stoss end which is a blunter and steeper than the other end
- The stoss end gets blunter due to pushing by the moving ice
- The other end is called as tail

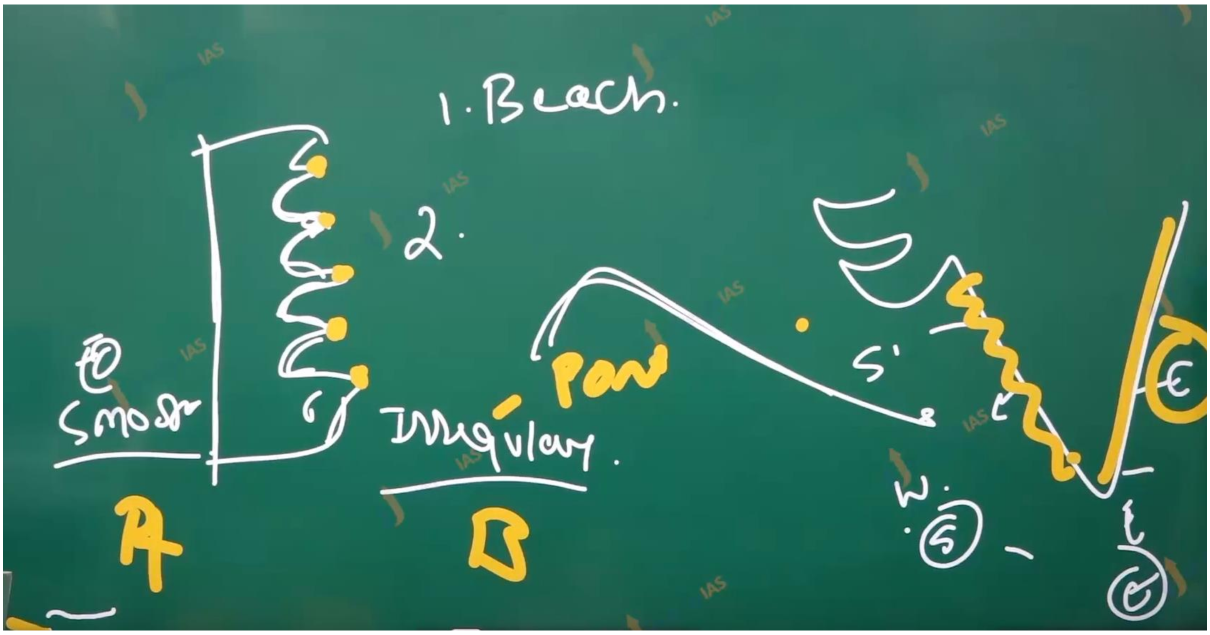


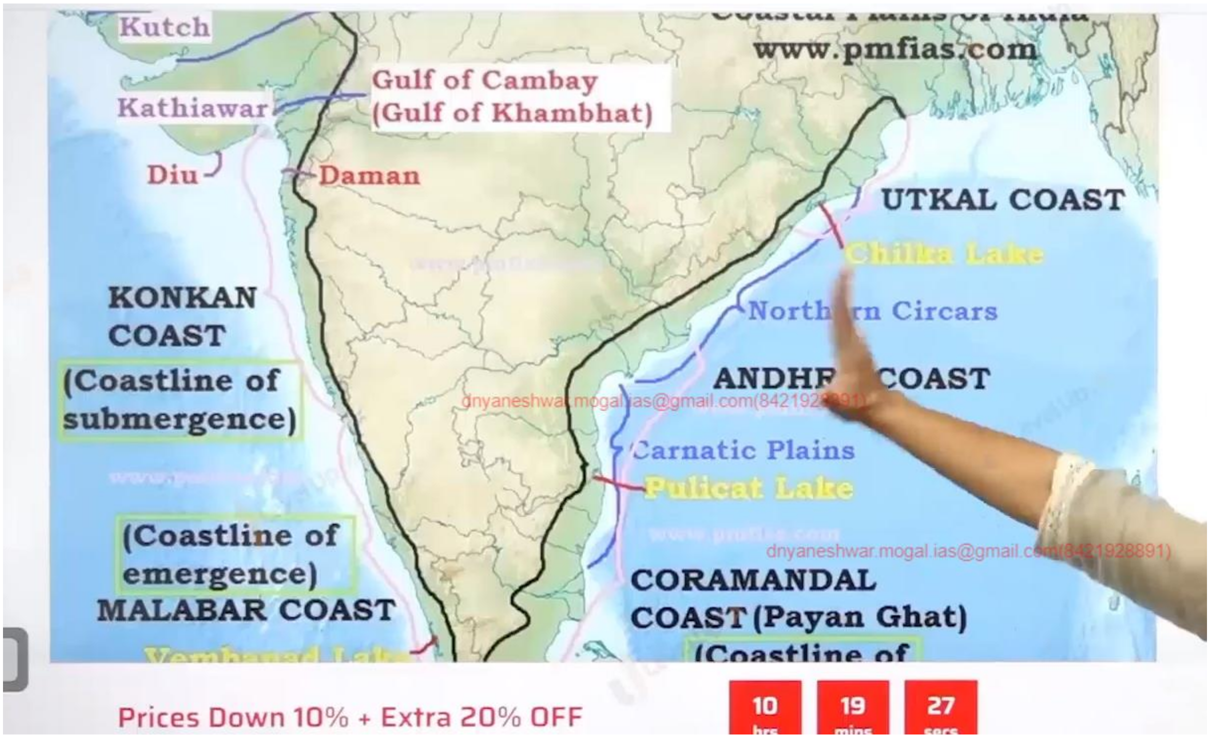
Waves



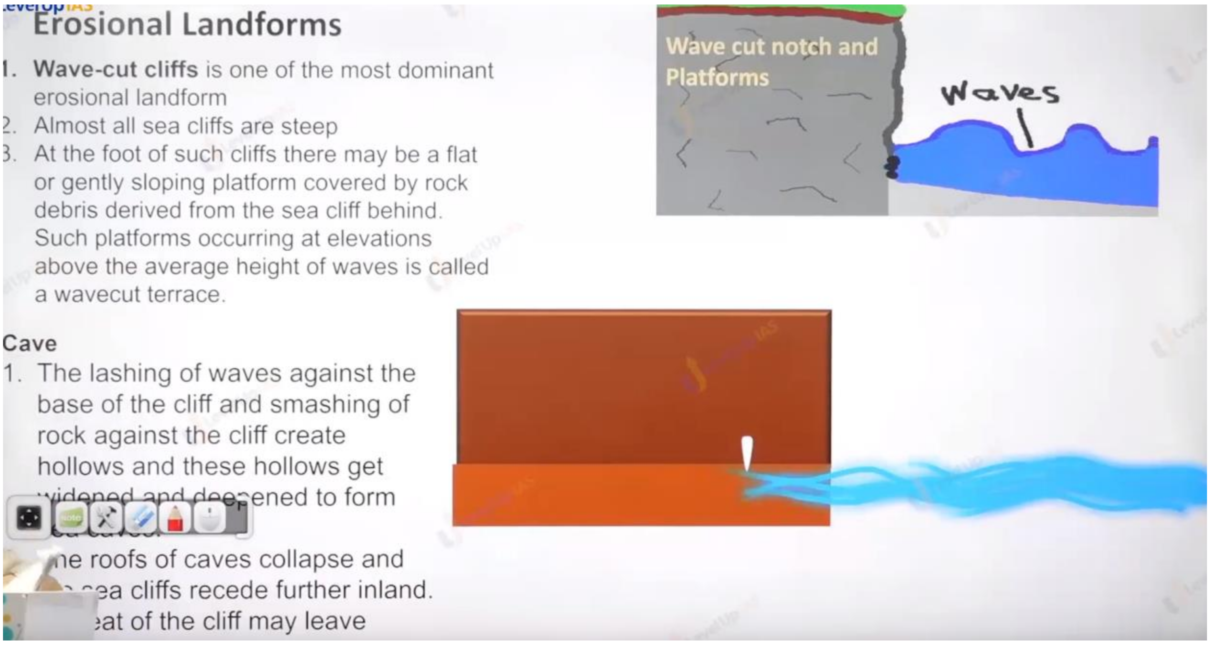
Waves:

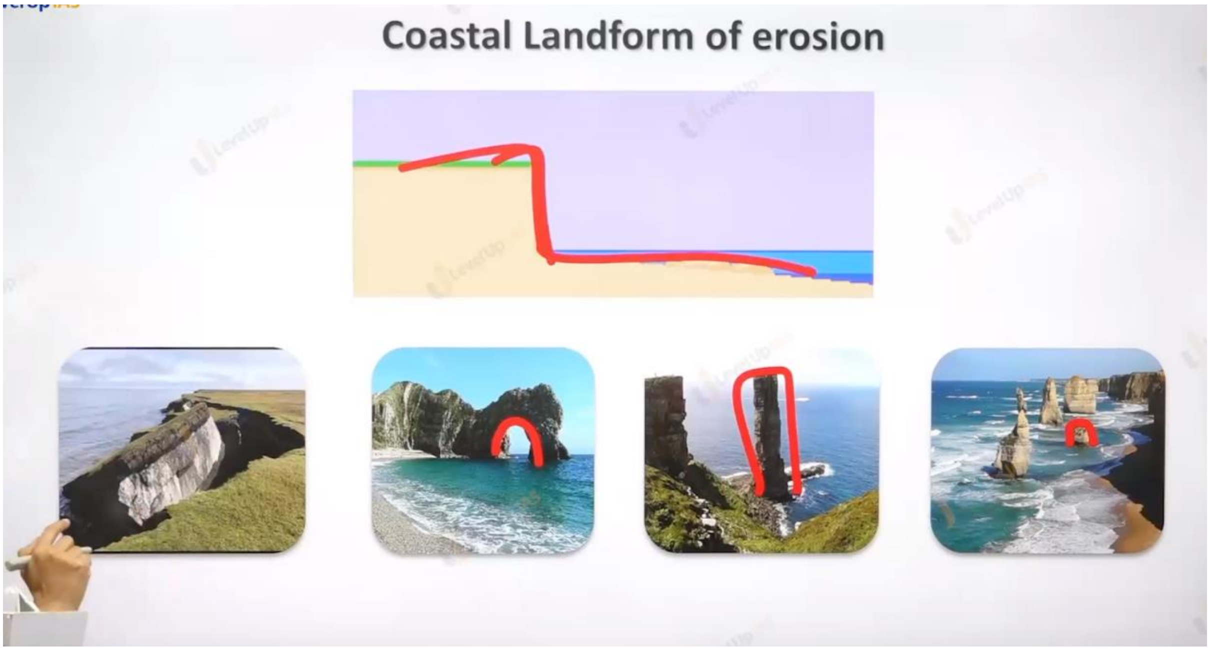
1. Waves are disturbances or ripples formed overseas surface as energy moves from one place to another
2. Do not involve movement of water particles it is the only the energy which moves.
3. Formed when producing force remains smaller than the resisting force due to which the energy is carry the water particle and and 's only successful in creating disturbances


1. Beach.
2.
Smooth
A
Irregular
B
Pond
5


Kutch
www.pmfias.com
Gulf of Cambay
(Gulf of Khambhat)
Kathiawar
Diu
Daman
UTKAL COAST
Chilka Lake
KONKAN
COAST
Northern Circars
(Coastline of
submergence)
ANDHRA COAST
Carnatic Plains
Pulicat Lake
(Coastline of
emergence)
MALABAR COAST
CORAMANDAL
COAST (Payan Ghat)
Prices Down 10% + Extra 20% OFF
10
19
27

Costal land forms

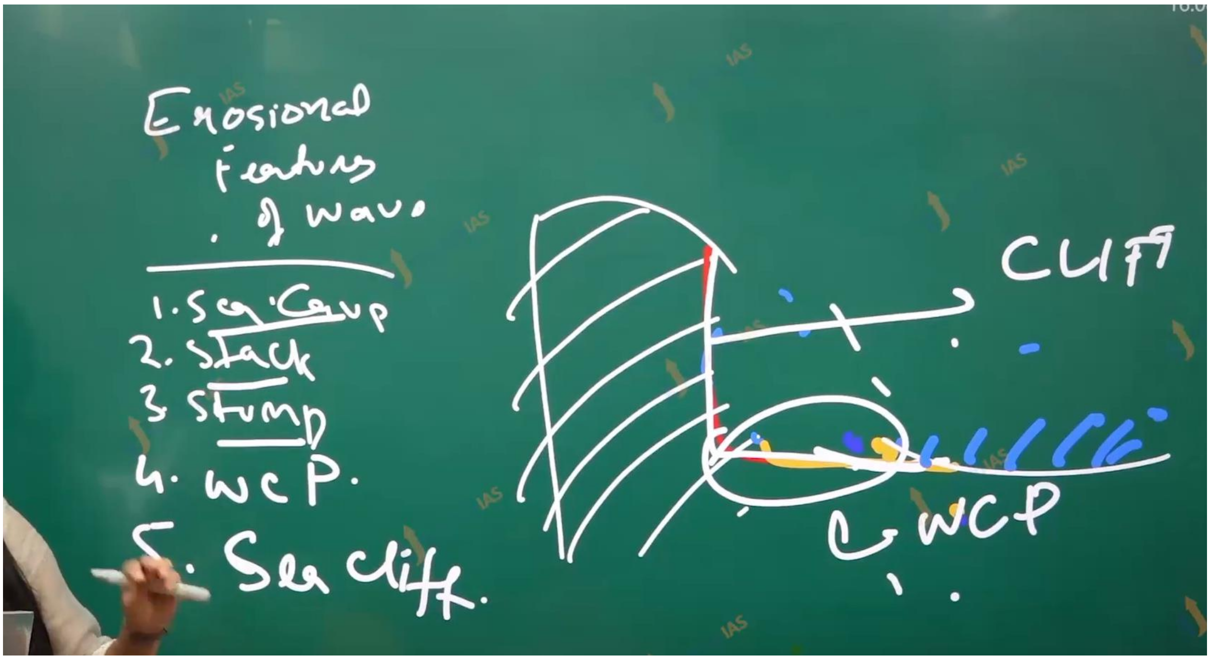



Coastal Landform of erosion


vel Up IAS
Coastal Landform of erosion


velUp IAS
Coastal Landform of erosion

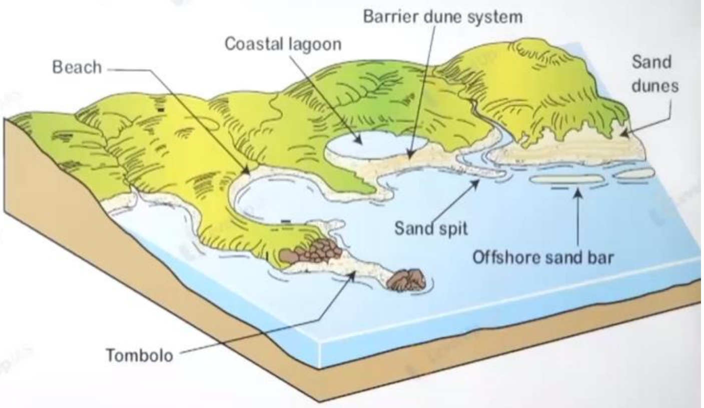

dnyaneshwar.mogal.ias@gmail.com(8421928891)
Off-shore Sand bars
Tombola
Beaches
Spits
hooks
Long shore current
dnyaneshwar.mogal.ias@gmail.com(8421928891)



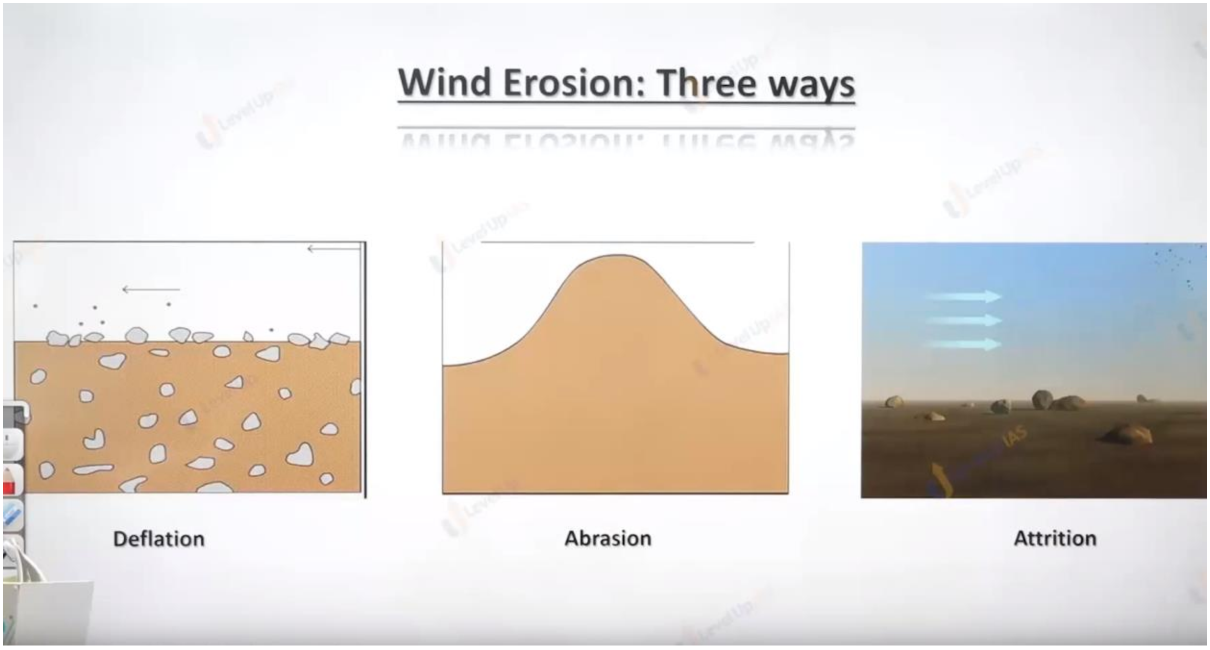
Formation of offshore Bars/ spits/ Tombola

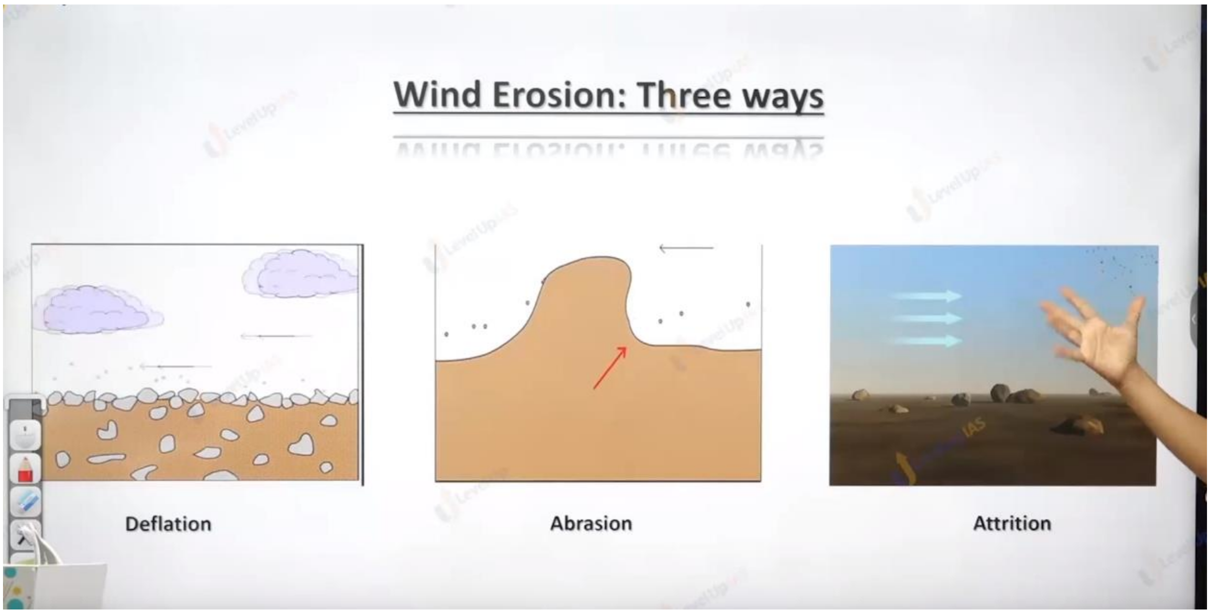
1. **Offshore Bar:** A ridge of sand and shingle formed in the sea in the off-shore zone lying approximately parallel to the coast is called an off-shore bar.
2. **Barrier Bar**: An off-shore bar which is exposed due to further addition of sand is termed a barrier bar.
3. **Spits:** When barrier bars get keyed up to one end of the bay when they are called spits

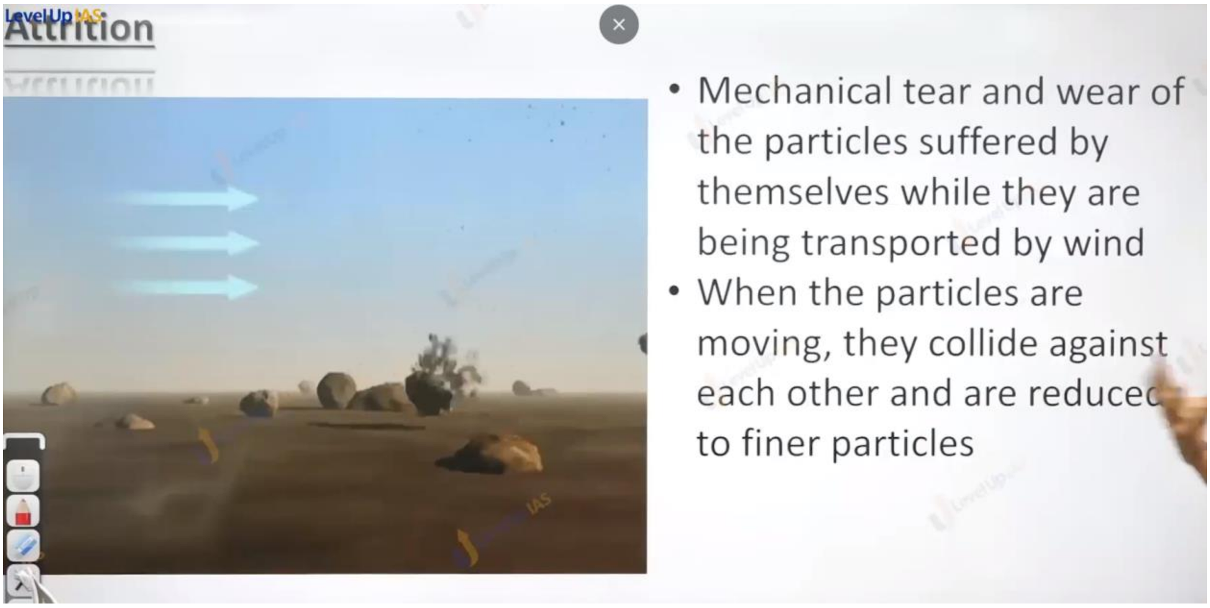
Barriers, bars and spits developed along a bay can gradually extend leaving only a small opening of the bay **into the sea forming a lagoon.** The lagoons get filled up gradually by sediment




Five types of deserts
Hamada
Hamada el Haroma in Sahara desert, Libya,
Mountain desert
Ahaggar mountains in Sahara desert, Algerian
Reg
Erg
Badland
Painted desert of Arizona
USA


Wind Erosion: Three ways
Deflation
Abrasion
Attrition


Wind Erosion: Three ways
Deflation
Abrasion
Attrition


LevelUp IAS
Attrition
• Mechanical tear and wear of the particles suffered by themselves while they are being transported by wind
• When the particles are moving, they collide against each other and are reduced to finer particles


Erosional Landforms
Mushroom Rocks
Deflation Hollows
Ventifacts and Dreikanters
Inselberg
Messa.. Butte.. Pinnacle

g

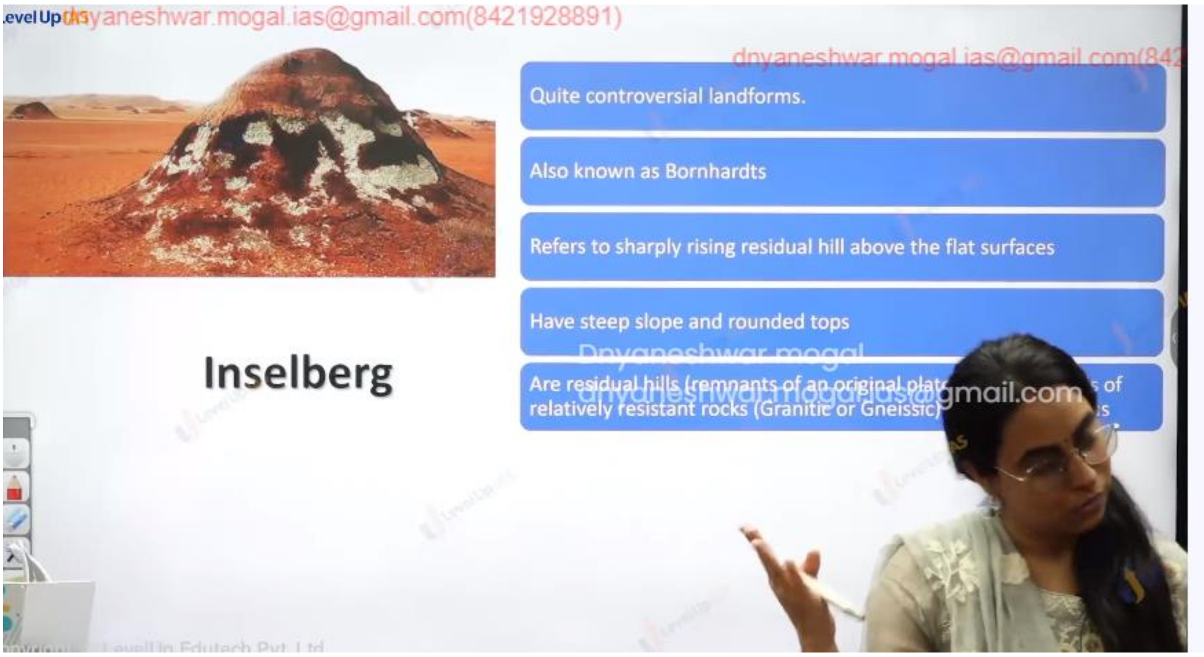
- Depressions in deserts formed due to removal of sands through the process of deflation
- The size of these enclosed
- depressions varies from smaller … like 'buffalo wallows' … e American Gre… lains to very large depres… such as great
- Qattara depression of Egypt

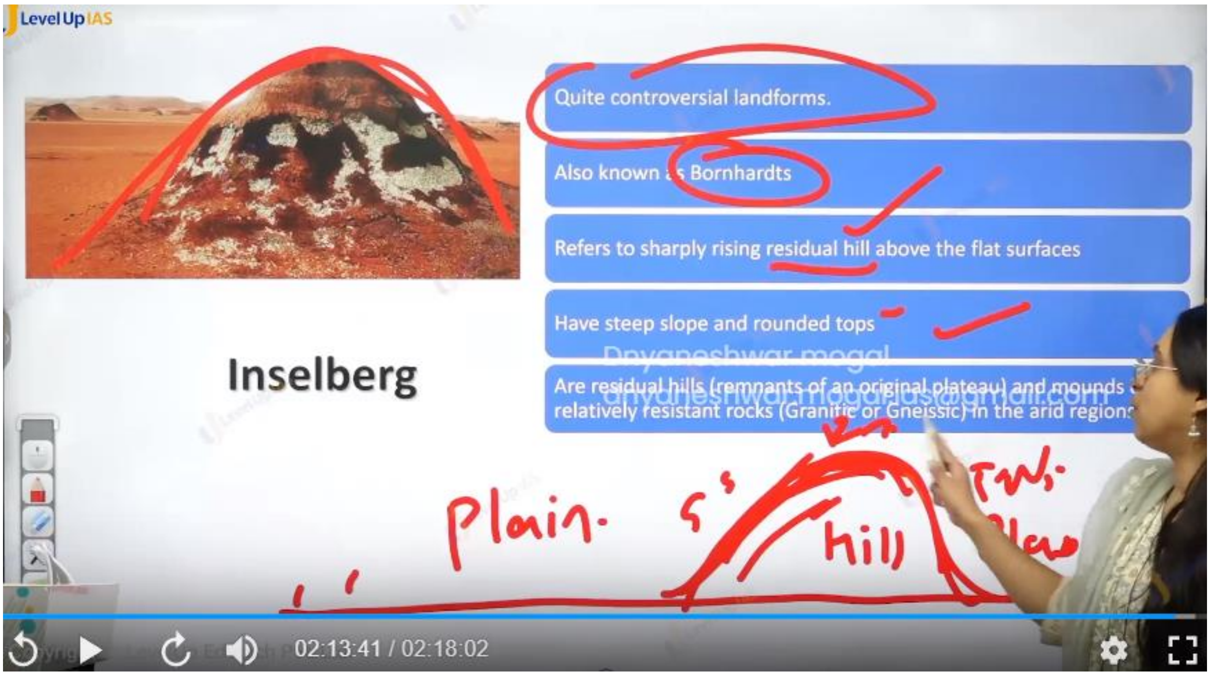
Deflation hollows or Deflation Basin or Blow …ts or desert Hollows

- Rocks having broad upper part and narrow base resembling an umbrella or mushroom
- These mushroom - shaped rocks are formed due to ABRASION
- Base of rock block is abraded vigorously from all sides because of variable directions of wind
- Active abrasion limited to 6 feet height from the ground while the upper part is least affected by abrasion.

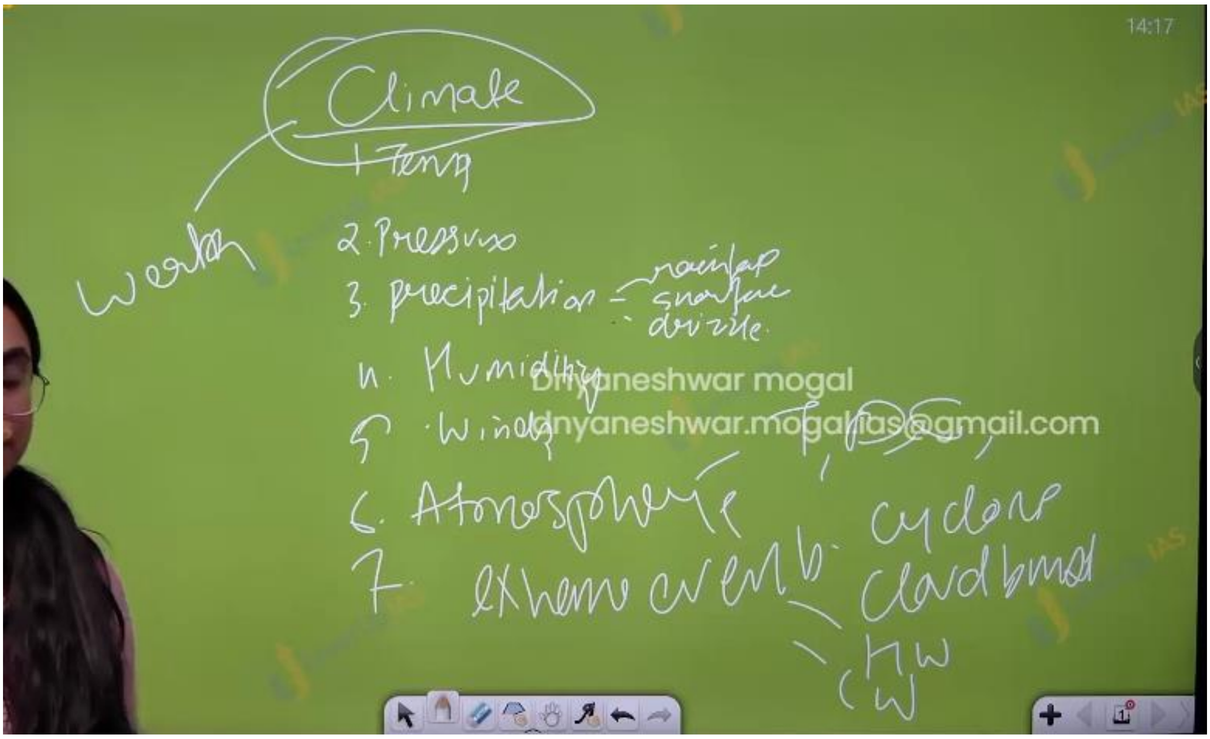
The Mushroom Rock

…E WHITE DESERT EGYPT


Level Up IAS
Inselberg
Quite controversial landforms.
Also known as Bornhardts
Refers to sharply rising residual hill above the flat surfaces
Have steep slope and rounded tops


Level Up IAS
Inselberg
Quite controversial landforms.
Also known as Bornhardts
Refers to sharply rising residual hill above the flat surfaces
Have steep slope and rounded tops
Are residual hills (remnants of an original plateau) and mounds
relatively resistant rocks (Granitic or Gneissic) in the arid region
Plain.
hill
02:13:41 / 02:18:02

r



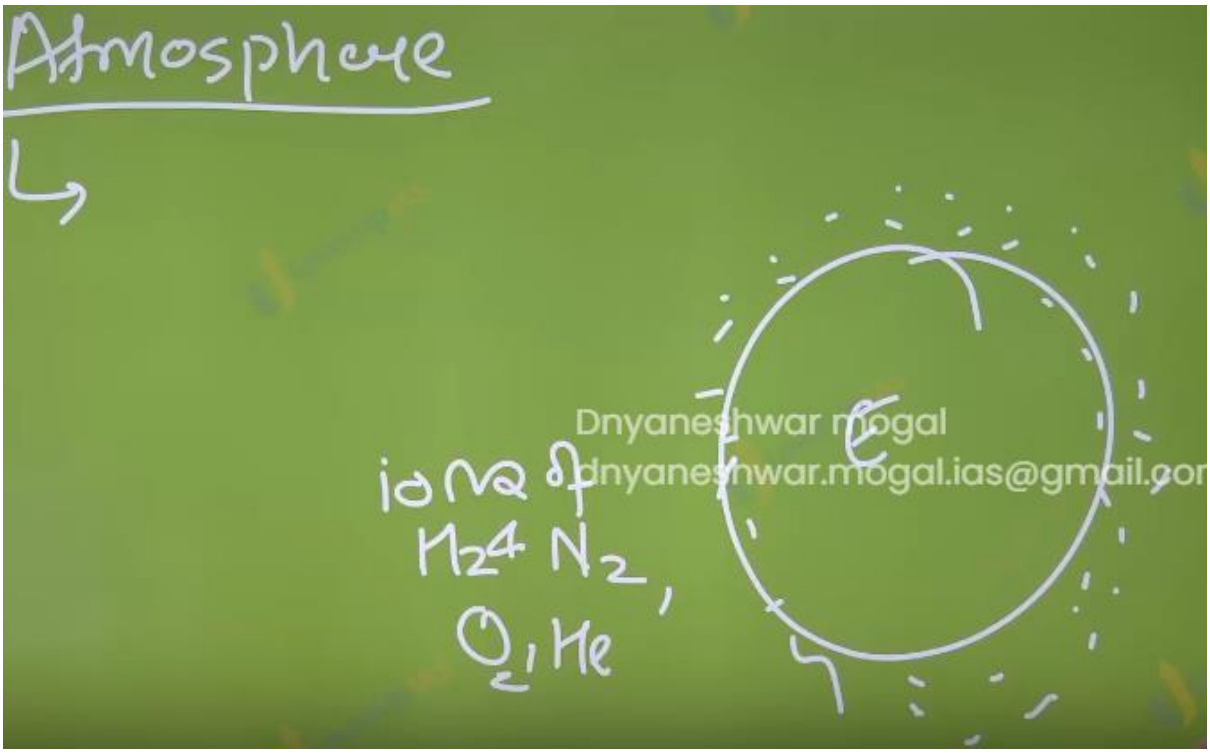
Climate

(14) Composition of Atmosphere and Layering

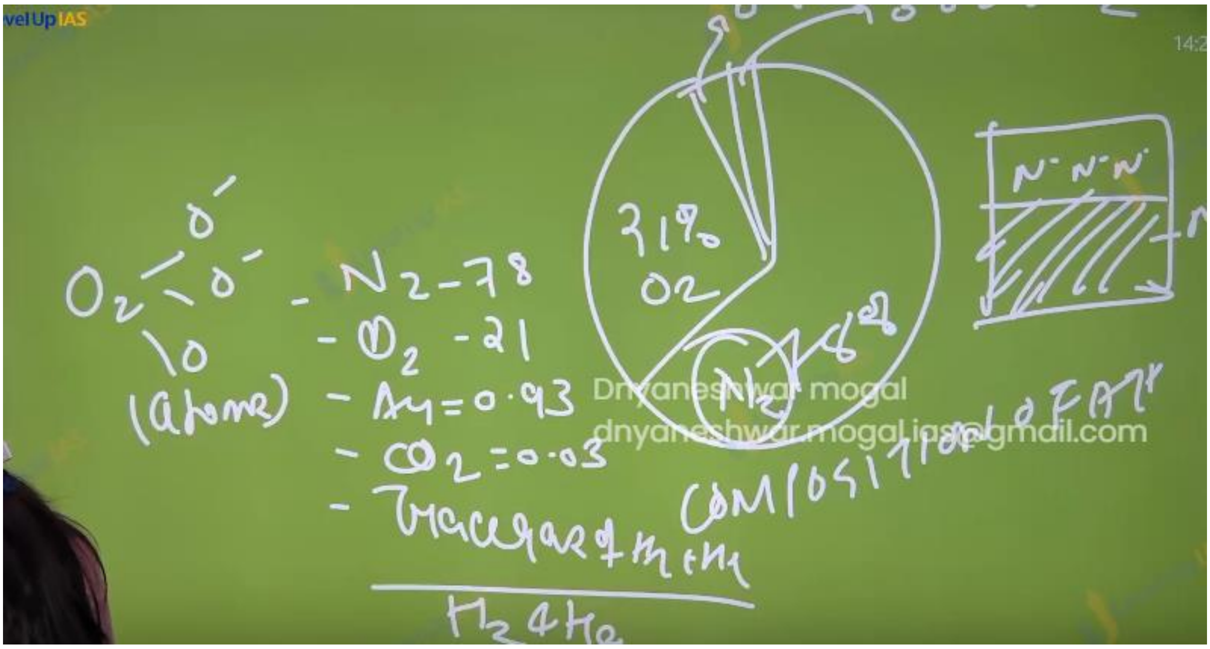
Climate :

1. Atmosphere.
2. Temperature
3. Pressure
4. Humidity
5. Precipitation ex rainfall, snowfall, grizzles.
6. Wind   ex sea wind , trade wind , monsoon wind,
7. Extreme event   cyclone, cloud bust, heat wave cold wave.

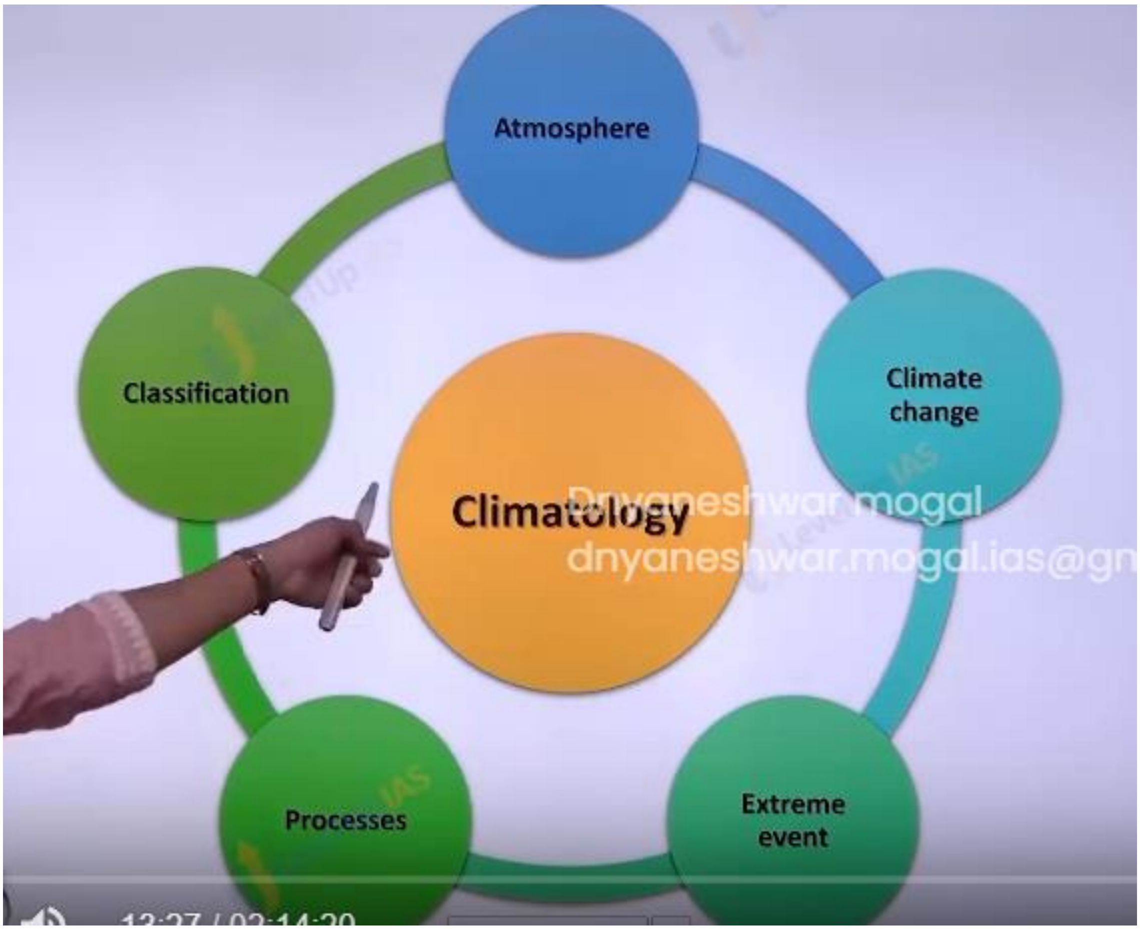
Atmosphere: blank of the gasses that surround the earth. It is composed of gassees dust particles water vapors pollutant, ions, nitrogen, oxygen. Helium.

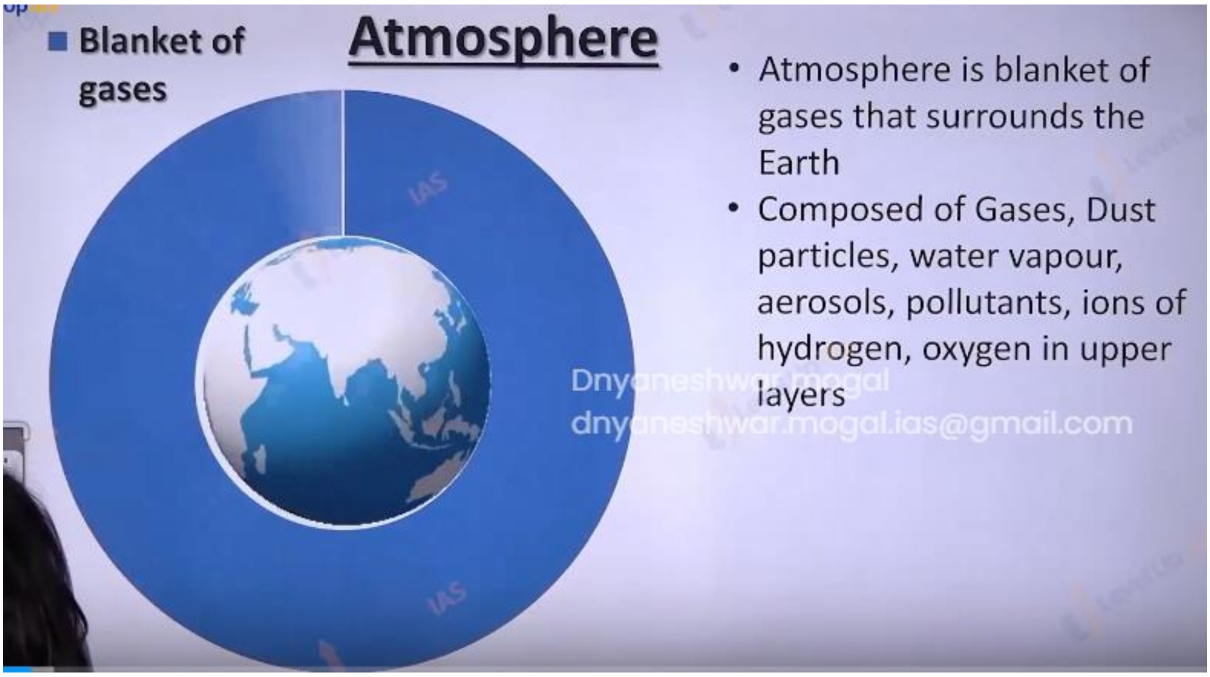


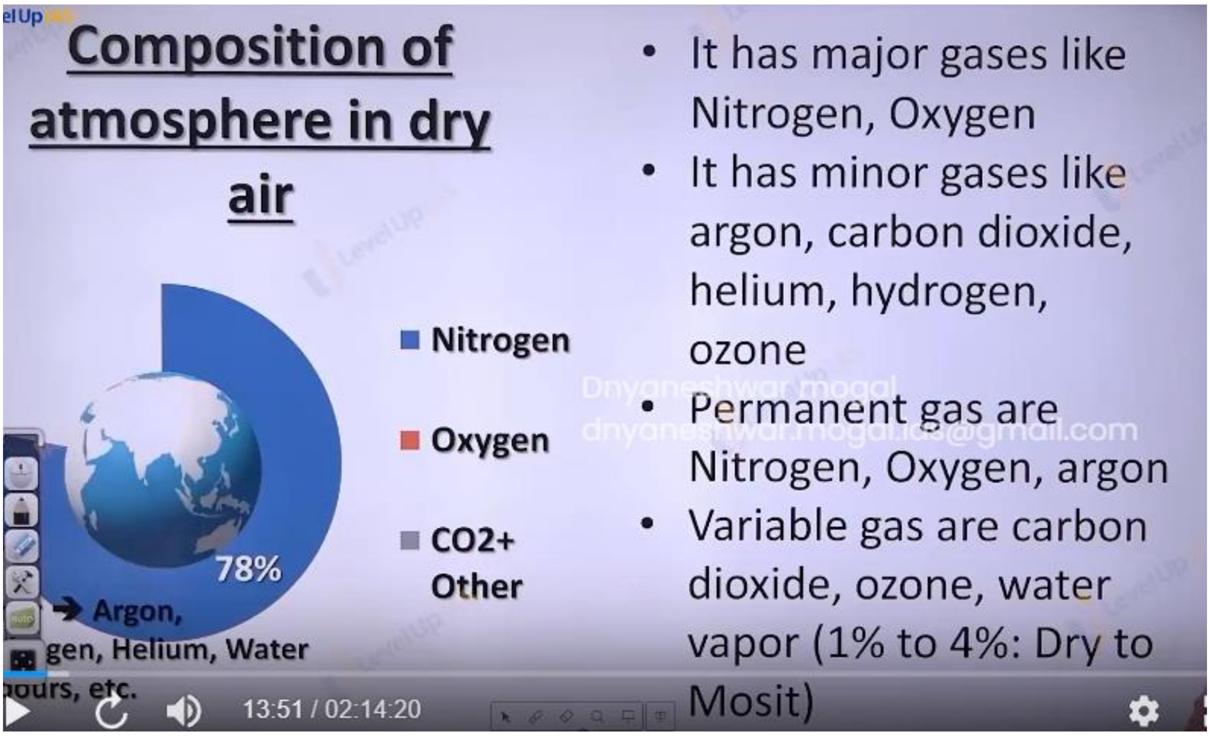
Composition of atmosphere:

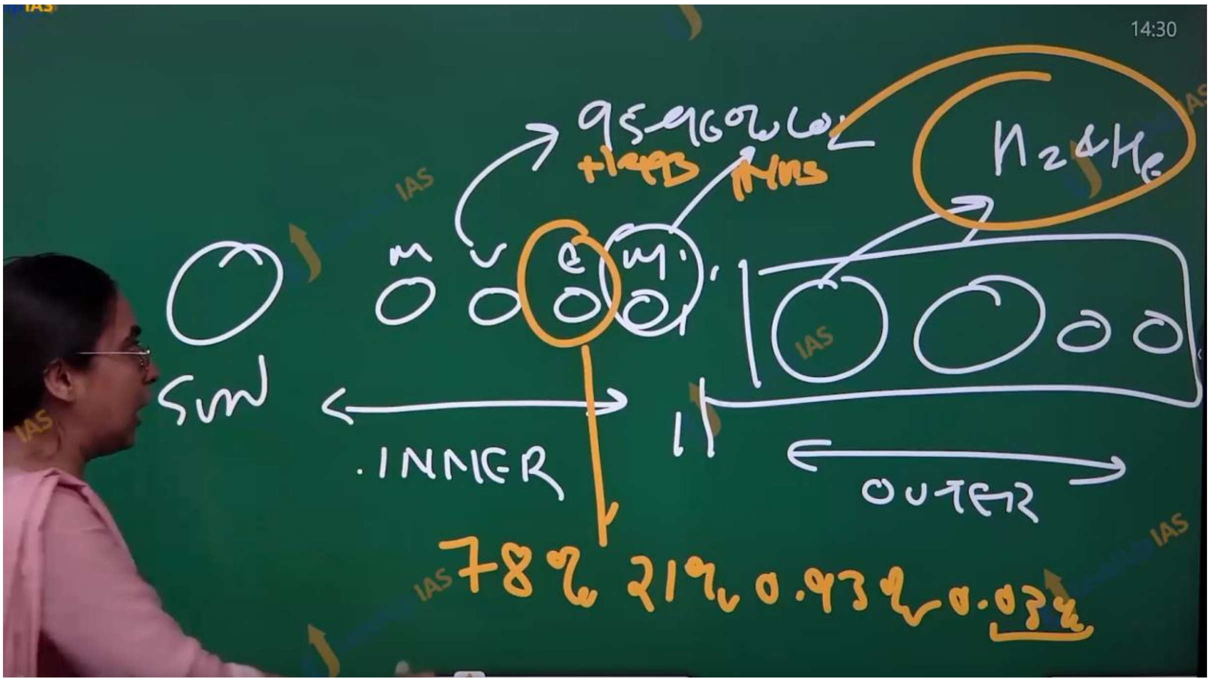


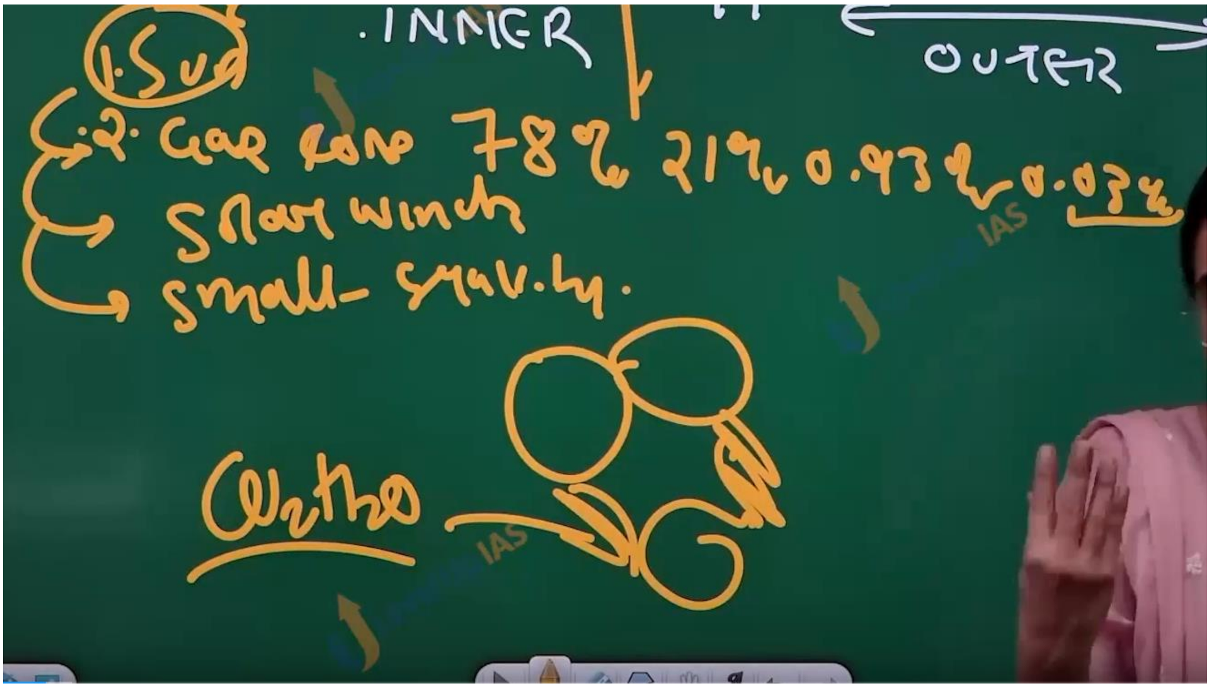
Element of climent.


Atmosphere
Classification
Climate change
Climatology
Processes
Extreme event
Dnyaneshwar mogal
dnyaneshwar.mogal.ias@gm


Atmosphere
Blanket of gases
• Atmosphere is blanket of gases that surrounds the Earth
• Composed of Gases, Dust particles, water vapour, aerosols, pollutants, ions of hydrogen, oxygen in upper layers
Dnyaneshwar mogal
dnyaneshwar.mogal.ias@gmail.com


Composition of atmosphere in dry air
Nitrogen
Oxygen
CO2+ Other
78%
Argon,
gen, Helium, Water
ours, etc.
It has major gases like Nitrogen, Oxygen
It has minor gases like argon, carbon dioxide, helium, hydrogen, ozone
Permanent gas are Nitrogen, Oxygen, argon
Variable gas are carbon dioxide, ozone, water vapor (1% to 4%: Dry to Mosit)
Dnyaneshwar mogal
dnyaneshwar.mogal.ias@gmail.com
13:51 / 02:14:20


14:30
95-96% CO2
Sun
INNER
OUTER
78% 21% 0.93% 0.03%

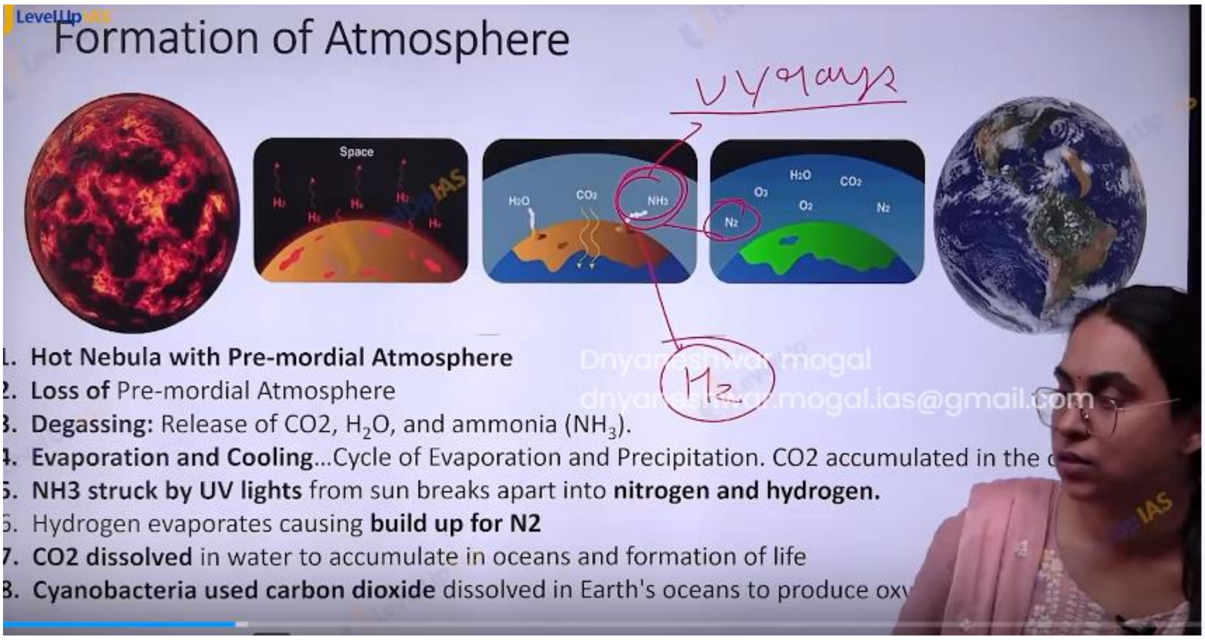


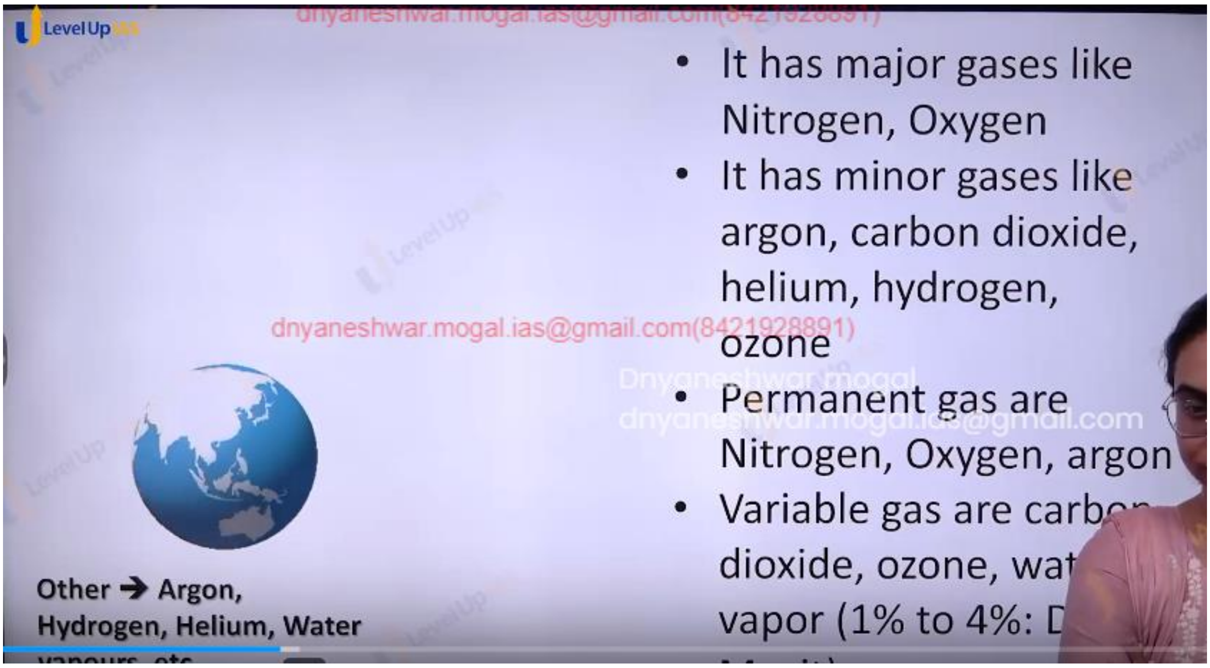
1. For most inner planets hydrogen and helium remain in the gases form due to high temperature. And continuous event of solar flare blows the gas away to outer planet.
2. Earth is at a suitable distance here carbon dioxide get the chance to settle down in the ocean and from co2h2o also called carbonic acid.

Phases of evolution of atmosphere:

1. Earth been the hot nebula with pre mordial atmosphere composed of hydrogen and helium
2. Loss of pre mordial atmosphere because of effect of solar wind.
3. The degassing stage. And the release of carbon dioxide, water vapour and ammonia
4. The cycle of evaporation cooling of water vapor condensation and precipitation in this process the water droplet the dissolve the co2 which accumulates in form of ocean water.

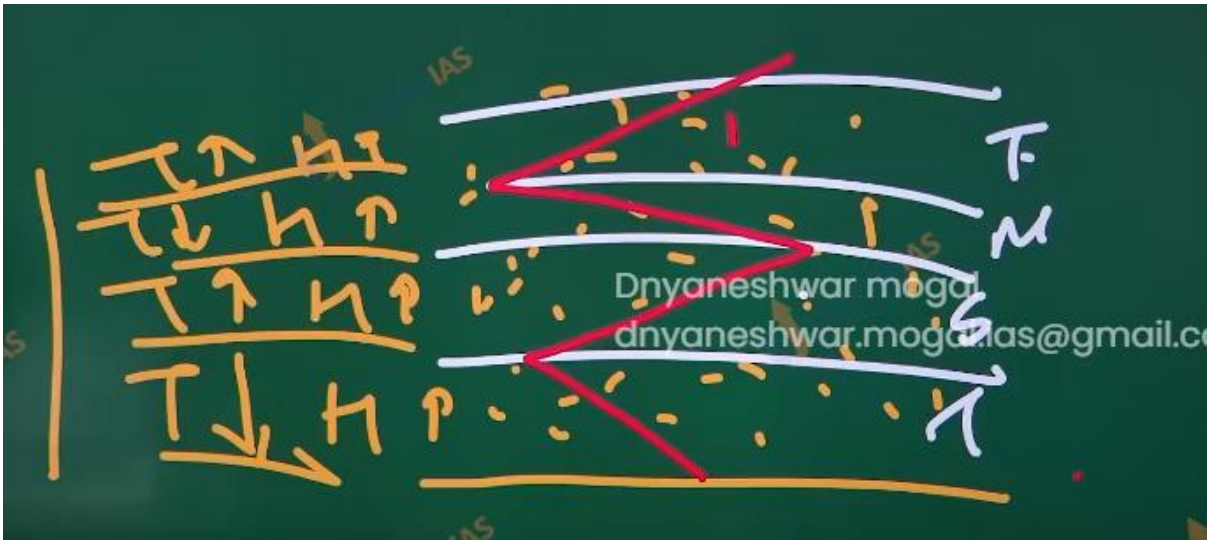
5. Ammonia strikes by the UV rays to split into nitrogen and hydrogen.
6. Hydrogen gas evaporates and nitrogen continues to build up.
7. Stage of formation of oxygen; in the ocean with the formation of life like cyno bacteria co2 is utilized and the oxygen is released as by product once the ocean water is saturated as a biproduct oxygen becomes the part of atmosphere.

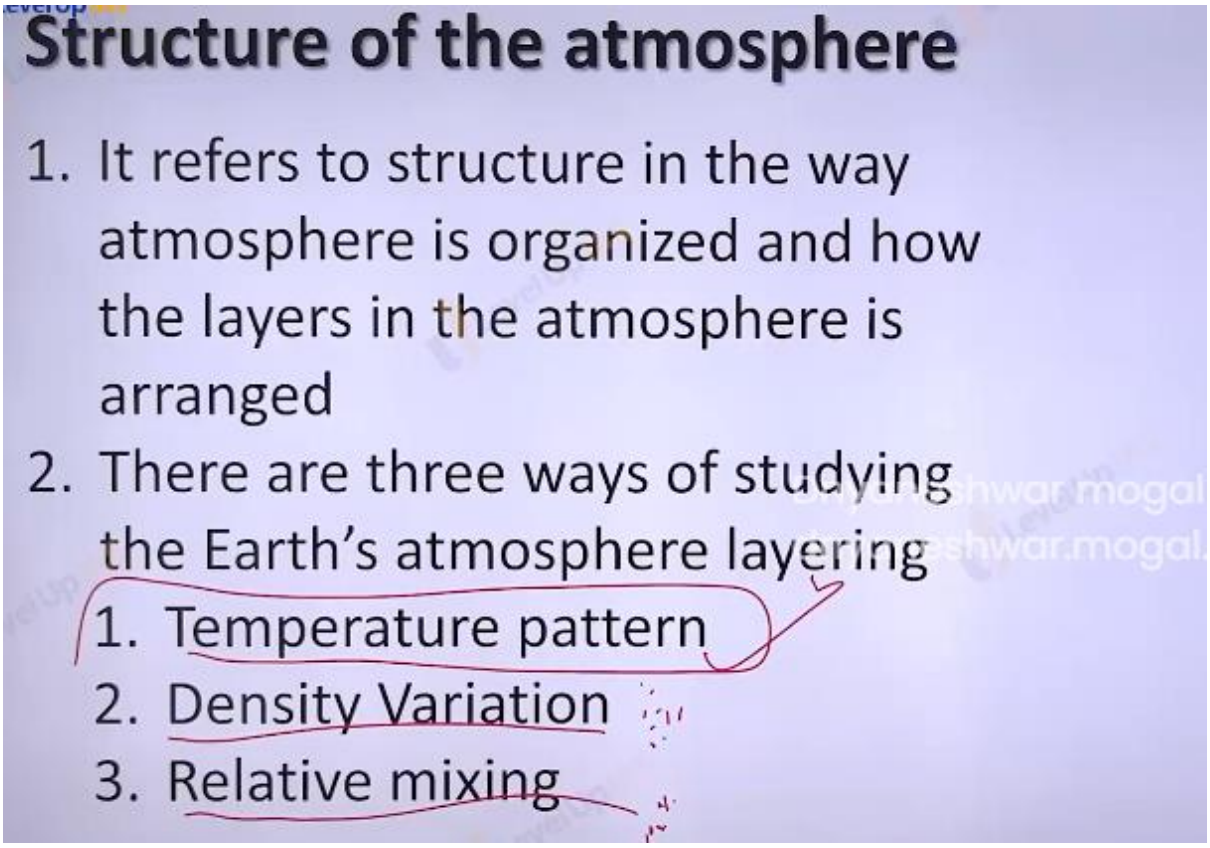




1. Nitrogen oxygen and argon do not directly play a important role in various climatic processes like wind rain fall as they are not greenhouse gases.
2. Green house gases are those that observe the heat ex, methane, Co2 ozone. Etc.

3. Layer of atmosphere based the changes in the temperature.





Layering of the atmosphere:

It refers to structure in the way atmosphere is organized how the layer are arranged. There can be 3 ways of layering

On the basis of temperature pattern.

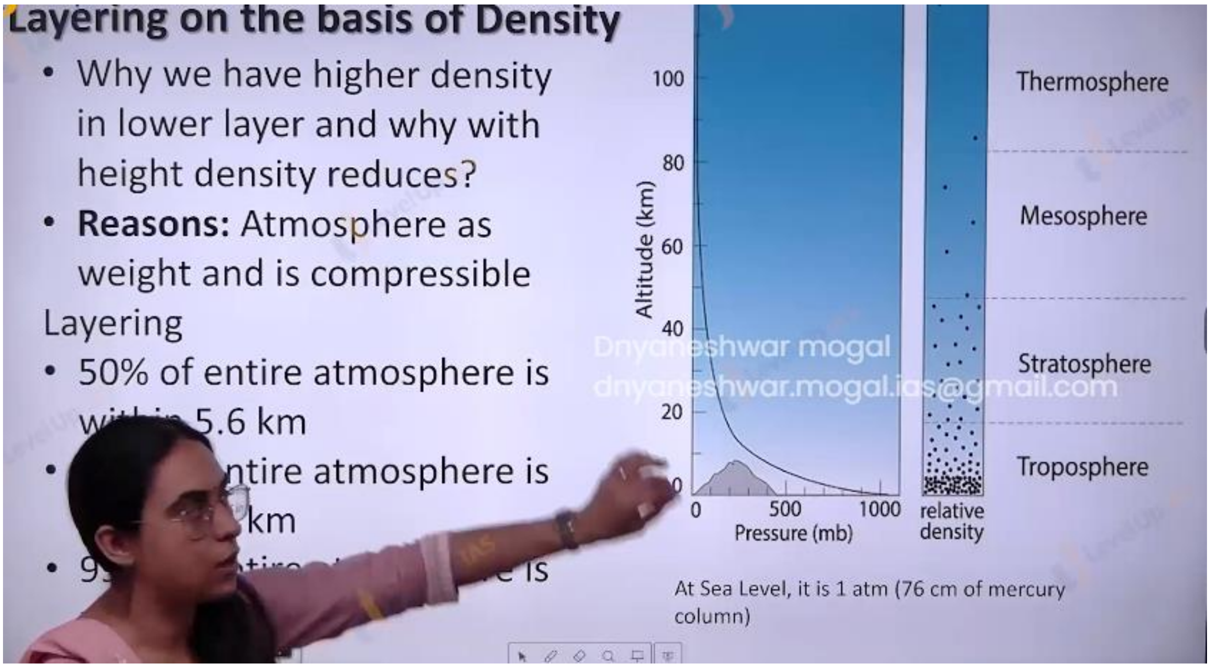
On the basis of relative mixing

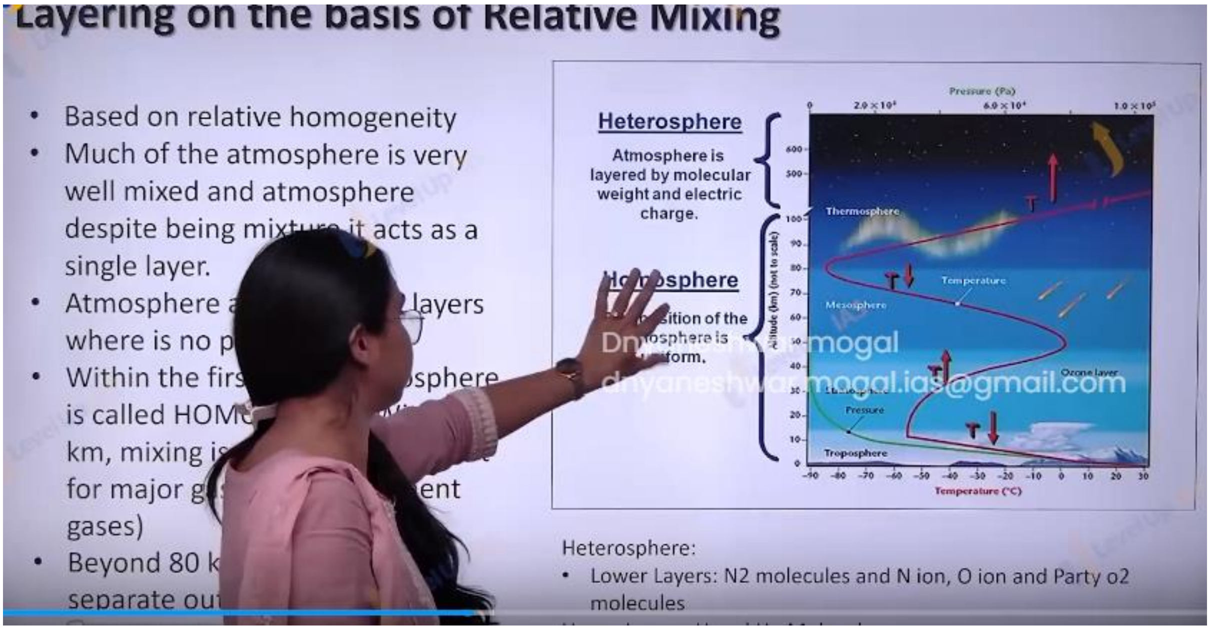
On the basis of density variation.

Atmospheric layering on the basis of density variation.

---

1. Atmosphere has weight and it is compressible. Nealy 50% of atmosphere within 5.6 km.
2. 90 % within the 16km
3. 99% atmosphere are within the 30km.





Layering on the basis of relative mixing:

1. Much of the atmosphere is very well mixed despite being the mixture of gases it act as one single layer of gas. This is seen in the first 80 km it is called in homosphere.
2. Above 80 km the gases and iron are separated out on the basis of their weight. This is called as hetrosphere.
3. The lower layer have the nitrogen molecules and nitrogen iron along with oxygen ions and oxygen molecules while upper layers have hydrogen and helium molecules.
4. Layer on the basis of temperature pattern.
5. The layers are troposphere, stratosphere, mesosphere, thermosphere, exosphere.
6. The temperature reduces with increasing in height in troposphere, increasing in stratosphere, reduces in the mesosphere, increases in thermosphere& exosphere.

Heating mechanisms in the atmosphere.

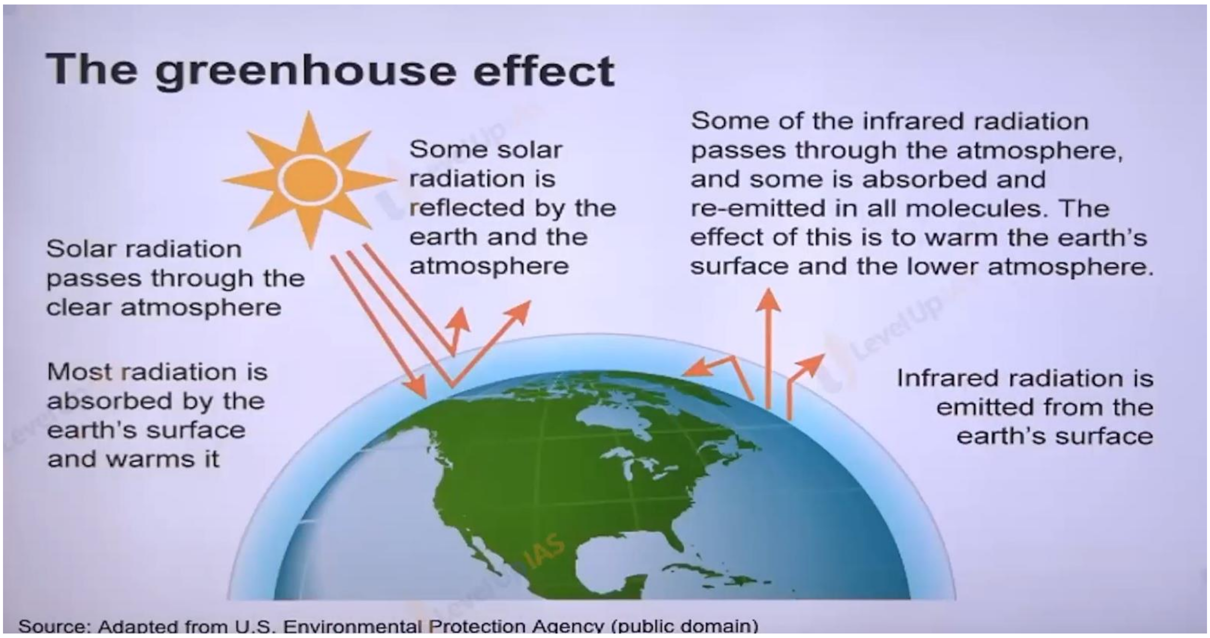
1. Atmosphere that surrounds the earth surface is heated in the two ways
2. observation of the UV rays by the ozone layer in troposphere
3. observation of the long wave radiation from the earth surface by the greenhouse gases. (long wave terrestrial radiation infra ray radiation)
4. Gamma x ultra visible infra micro radio.

5. Greenhouse gases are transparent to incoming solar radiation but are opeck the outgoing terrestrial radiation. The lower atmosphere heated to process of conduction (transfer of heat) convection (lower surface heated first it will move to upper) and advection (heat is travelling horizontally heatwave).

Green house gases

1. Carben dioxide
2. Methane
3. Nitrogen
4. Nitrous oxide
5. Hydrofluorocarbons
6. Perfluorocarbon
7. Sulfur hexafluoride

8. Nitrogen trifluoride.
9. Ozone
10. HCFC.



Q. which of the following is correct.

---

1. The amount of insulation received at the equator is roughly 10 time more than pole.
2. Infrared radiation constitutes 2/3 rd of the insulation.
3. Infrared radiation is largely observed by the water vapor that is concentrated at the lower atmosphere.
4. infrared radiations are part of visible spectrum of electromagnetic radiation

---

The answer is 3. Because UVI Ultra and visible spectrum contains 63 percent and infrared contains 37 percent not approx. 1/3 b is wrong

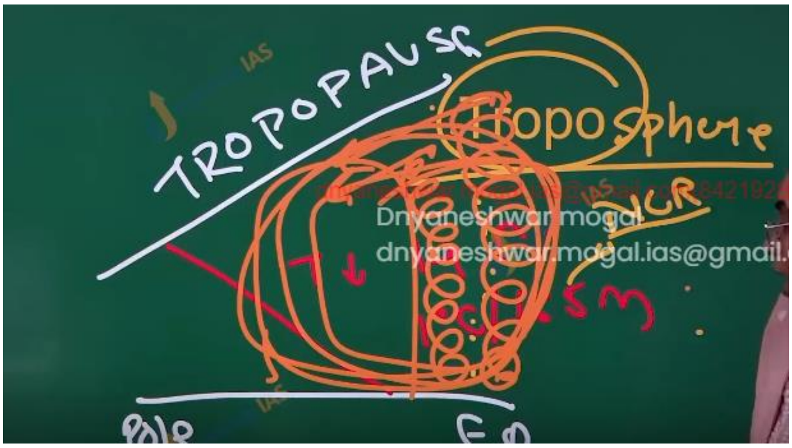
the insulation is more at the equator but not 10 times.

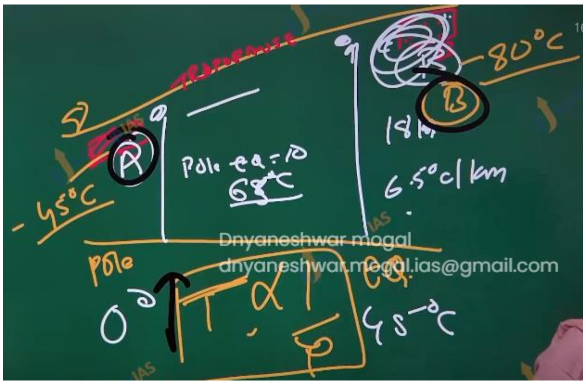
Layer of atmosphere on the basis of temperature pattern.

Troposphere.

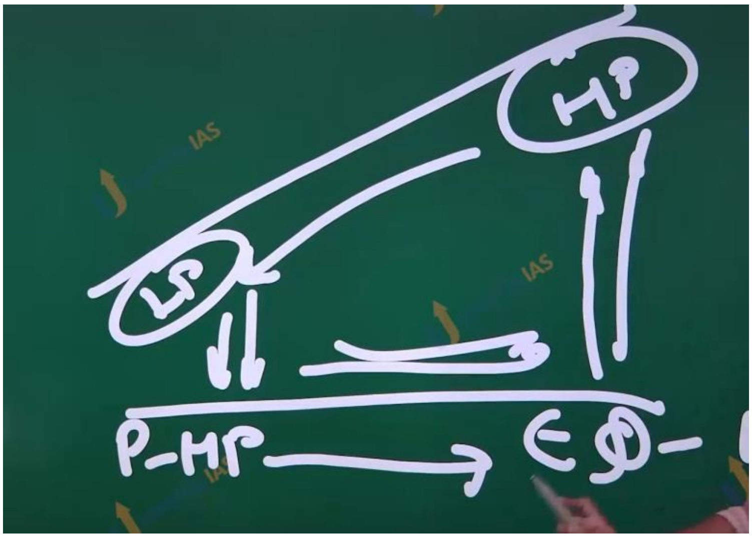
---

1. The temperature reduces with the increase in the height as troposphere is heated from the below out going long wave terrestrial radiation.
2. The rate of cooling is 1-degree Celsius per 165 m or 6.4 degrees Celsius per km this is called as normal lapse rate
3. The height of troposphere is 8 km at pole and 18 km at the equator, the average height is 13 km.
4. Thickness is greater at the equator because heat is transported to greater height by the conventional air currents, the air packet rises up to great height before it starts turning away. Troposphere word is derived from troop which means turning away.
5. The troposphere is also called convective reason, since all the convection is stop at tropopause.

6. 

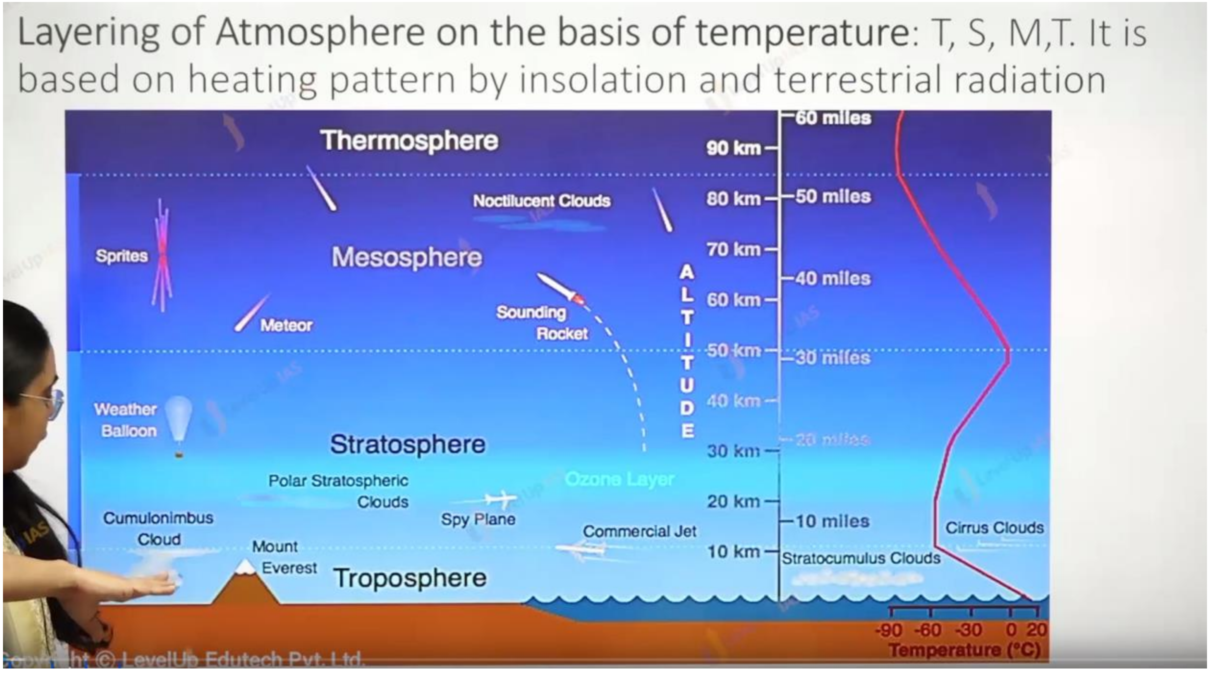
7. A
8. Troposphere experience fall in temperature the temperature Is -80 degrees Celsius at the equator at the tropopause. While at pole temperature is -45 degree Celsius at the tropopause.


9.
10. A
11. Meteorologically this most significant layer as the entire atmosphere and near all the weather phenomena like cyclone, rainfall, fog, Hellstrom, are confined to this layer.
12. Above the troposphere the temperature is constant in tropopause, therefore tropopause acts as lid and holds all the circulation within the tropopause.
13. All the Jet planes and aero planes fly in the tropopause in the lower stratosphere.

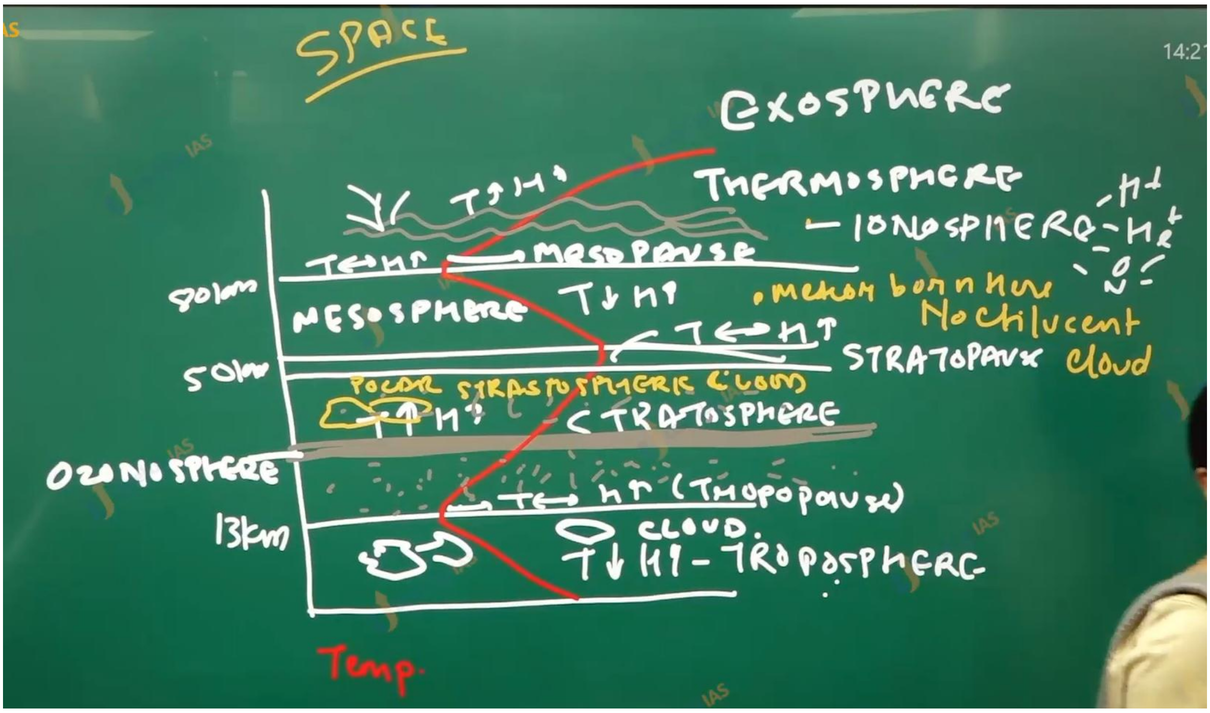


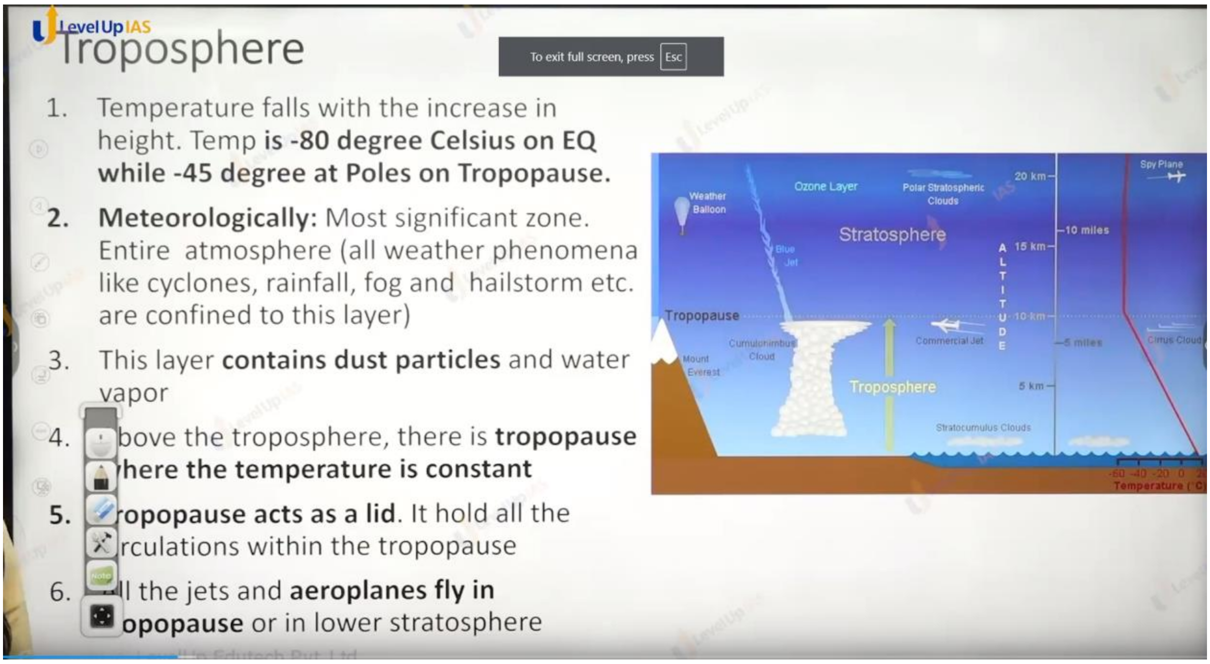
(15) Atmosphere Layering and Heat Budget

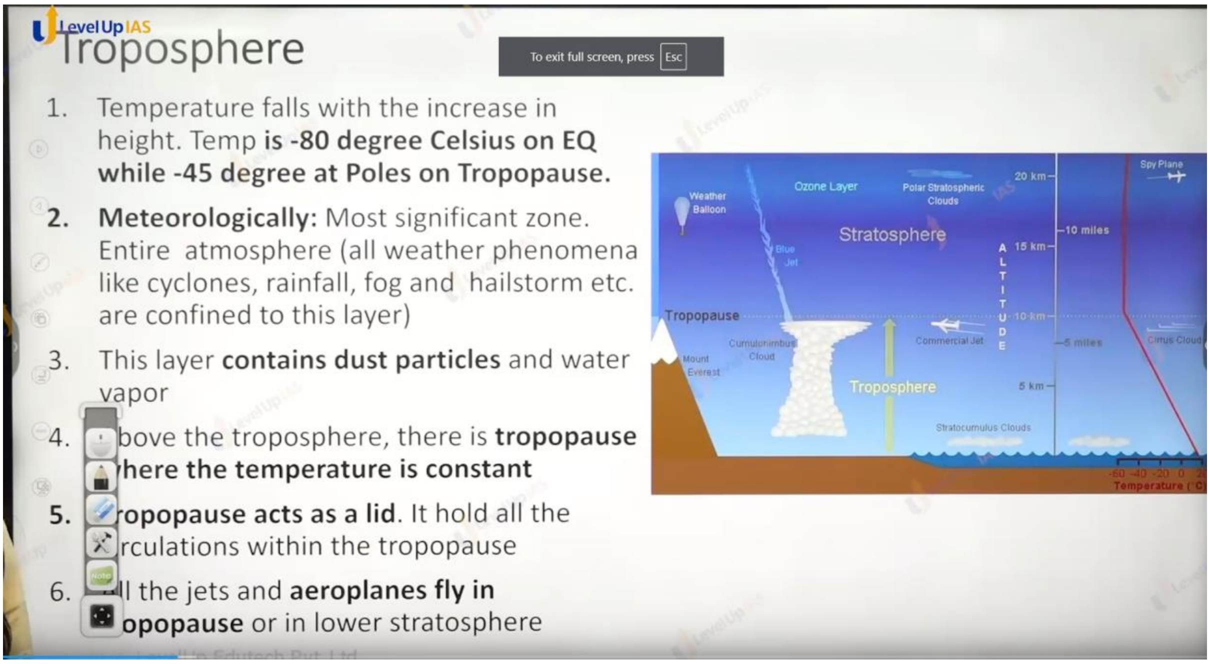
This

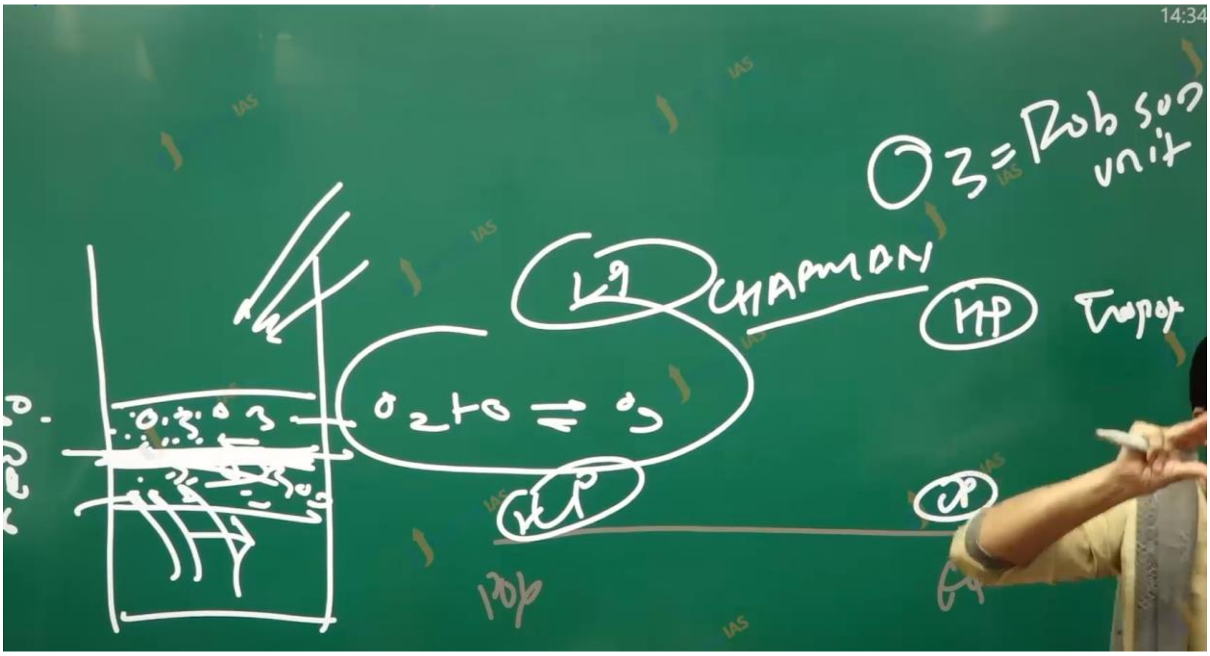


PV/T is not constant in the atmosphere PV/T is constant in the close system or close glass. In the real world pt is constant in the p is inversely proportional to temperature.









Stratosphere:

1. It is layer above the troposphere
2. Temperature increases with increase in the height
3. temperature reaches zero degree Celsius at 50 km. rise of temperature is due to present of ozone which observe infrared ray incoming and reflecting from surfaces.
4. This layer is almost free from the cloud and therefore the Aeroplan fly in this layer. (plane can also fly in the tropopause)

5. In the stratopause the temperature is constant with increasing in height.
6. Ozone molecules and Ozone gas present throughout the stratosphere that is 10 km to 50 km but there is buildup of Ozone between 20-30 km as UV rays' tents to split the ozone molecules up from 50 km to 30 km, and the terrestrial infrared radiation split the ozone molecules from the lowest stratosphere up to 20km.

Mesosphere:

1. It lies above the stratosphere and extends up to 80km.
2. Most meteors burn in this layer on entering space. Cause friction.
3. The temperature drops with increasing altitude up to mesopause. The temperature is -45 degrees Celsius and the mesopause is the coldest layer in the atmosphere. As the air is very cold even very scarce water vapor can condense into mesospheric cloud called noctilucent cloud.

Thermosphere:

1. In this layer the temperature rises rapidly with increasing in height because of radiation from the sun. (heated from the above) its extents between 80 – 400 km. though the temperature is very high any object and person in this layer does feel the heat because the atmosphere is very rarefied that Is gas molecules are spaces 100km apar.
2. The international space station and satellites orbit in this layer. The auroras are observed in the lower part of this layer.
3. The Karman line is located at an altitude of 100 km and used to define boundary between the space and earth's atmosphere.

Ionosphere:

1. The ultraviolet and x ray solar radiation ionizes the atom and molecules creating the layer of ions called ionosphere.

2. Ionosphere is plays important role for depleting the radio wave for communication
3. Discovery of the ionosphere was made by the author Kennelly and Oliver Heaviside who explain the transmission of radio wave signal around the curved earth surface therefore the ions rich layer also refers as Kennelly Heaviside layer.
4. Ionosphere has many layer D,E,F .
5. D is the lowest layer and it reflects the long wavelength radio wave and allows short and medium wavelength radio wave.
6. E layer reflects the medium wave radio wave and extents from 90-160 km
7. F layer reflects the short-wave radio wave and extends from 160-300 km

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Exosphere:

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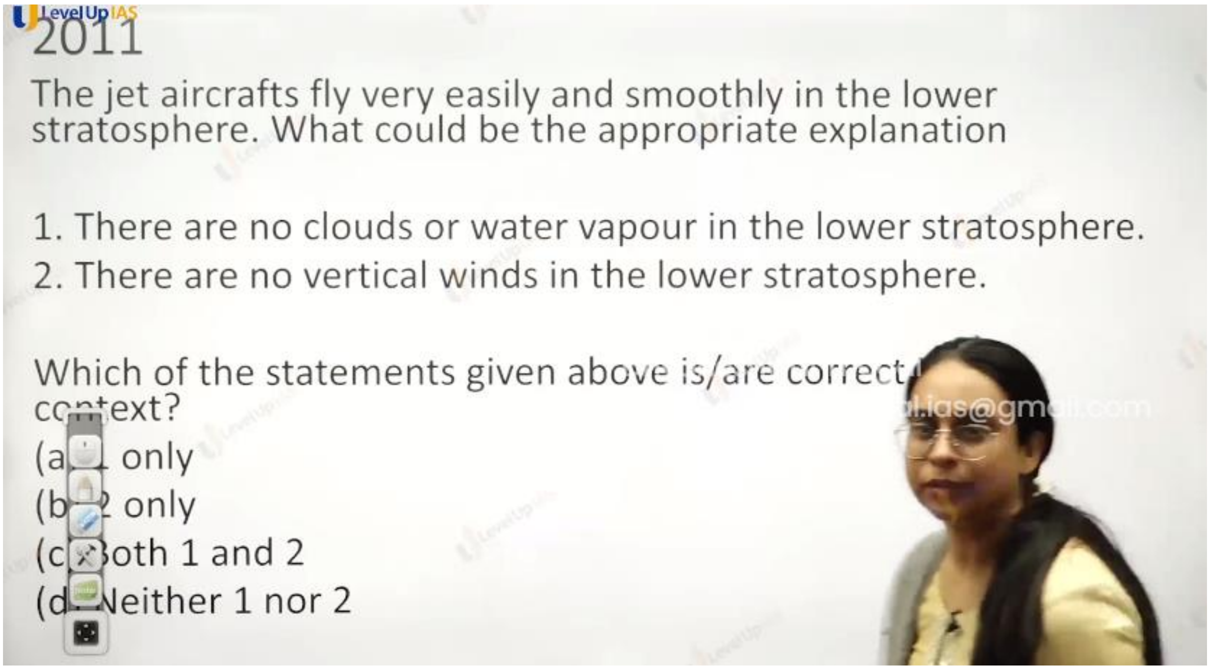
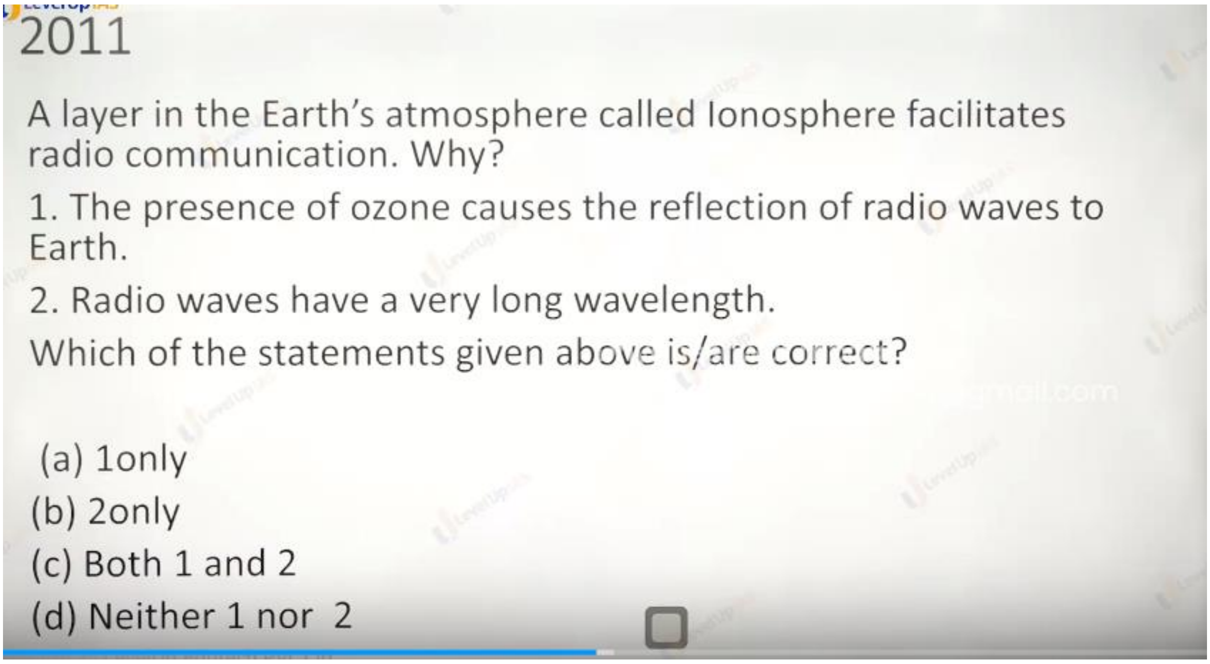
1. From the above the 400 km atmosphere is very rarefied
2. The temperature increases with increase in height.
3. It has light gases, and it considers space.



The answer is c because no clouds and water vapor less disturbance.

And in the troposphere, there is air conventional current so vertical wind meaning plane are stable and low fuel.



Option d

Ozone is present in the stratosphere.

For 2 short medium and long wavelength all are reflected. Cause of ions

Not because of the long wavelength of radio.

2012

Normally, the temperature decreases with the increase in height from the Earth's surface, because:

1. The atmosphere can be heated upwards only from the Earth's surface.
2. There is more moisture in the upper atmosphere.
3. The air is less dense in the upper atmosphere.

Select the correct answer using the codes given below:

(a) 1 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

Only c.

1 is obviously true and for 3 is somewhere true less molecule's small chance of getting heated.

2 moisture is less in the upper atmosphere as height increases.

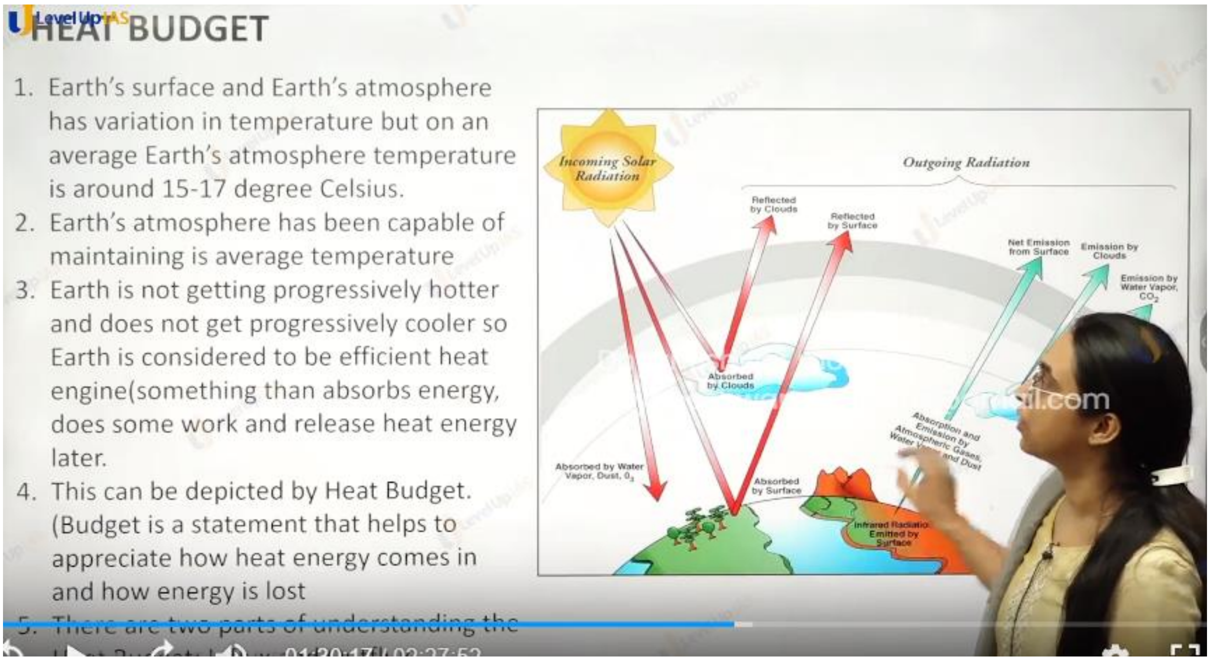
Mains Question: UPSC CSE 2022

- Troposphere is a very significant atmosphere layer that determines weather processes. How?

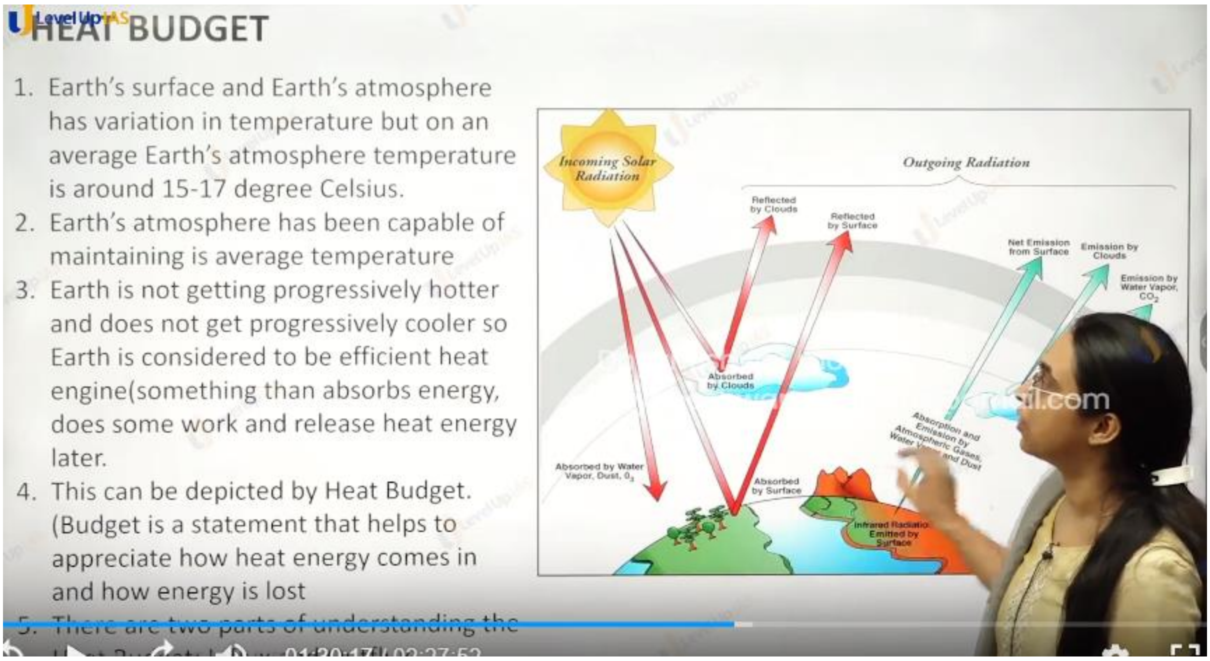
Temperature → pressure→ winds 1.

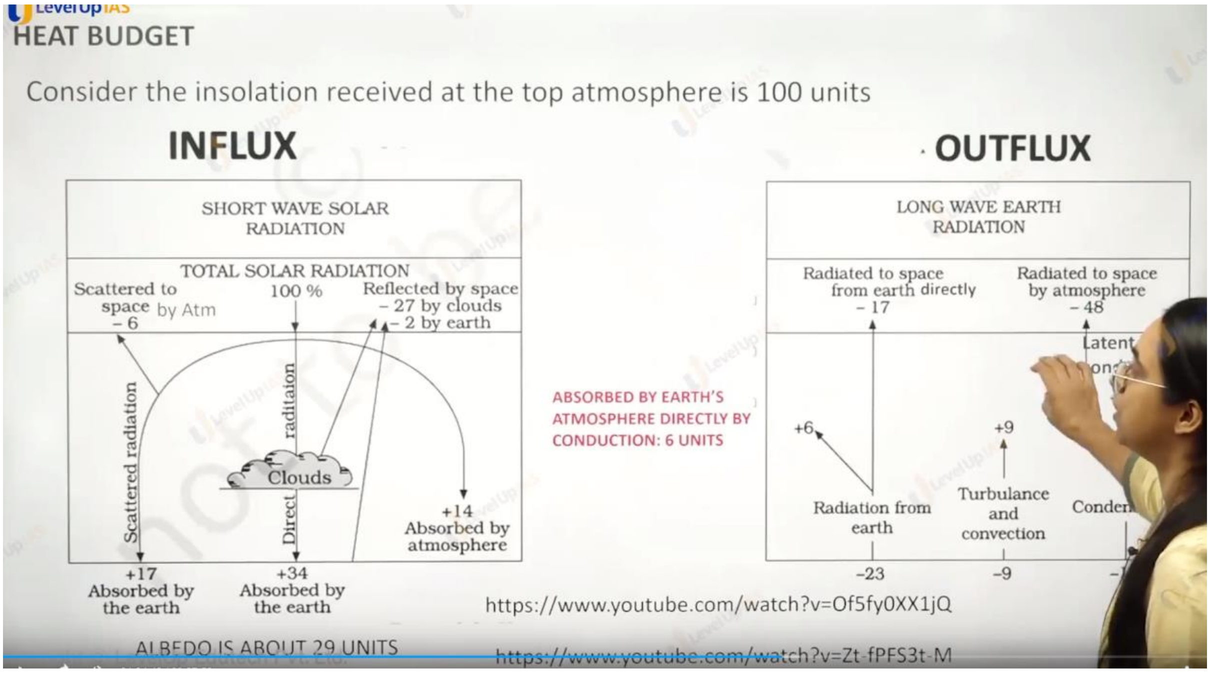
Climate : Heat budget , temperature pattern.

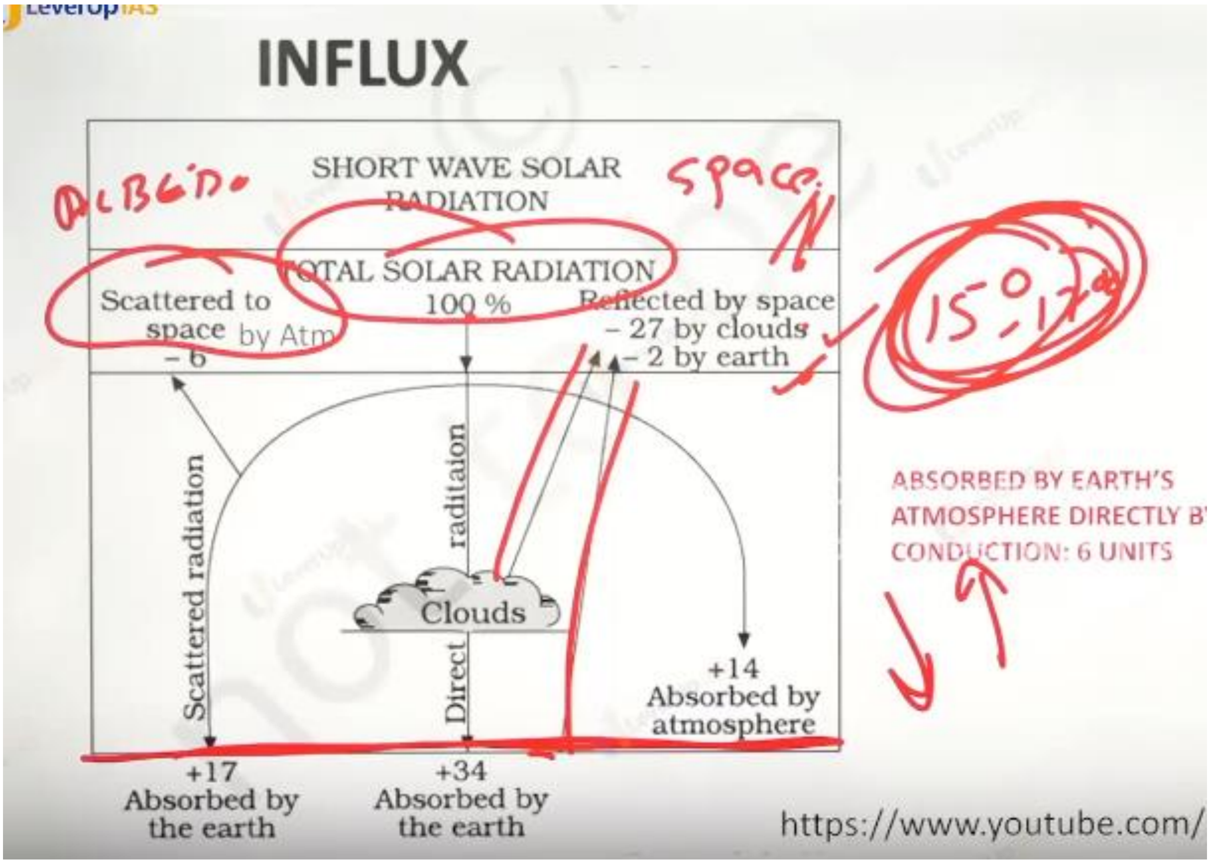
Heat budget.

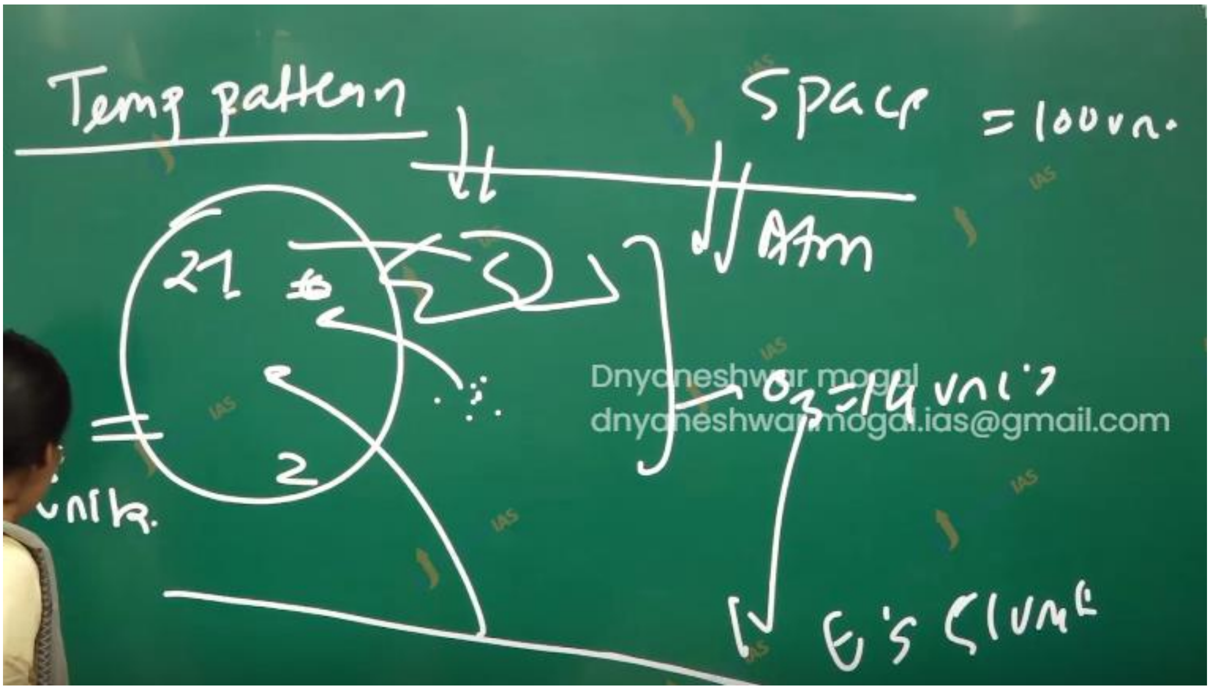


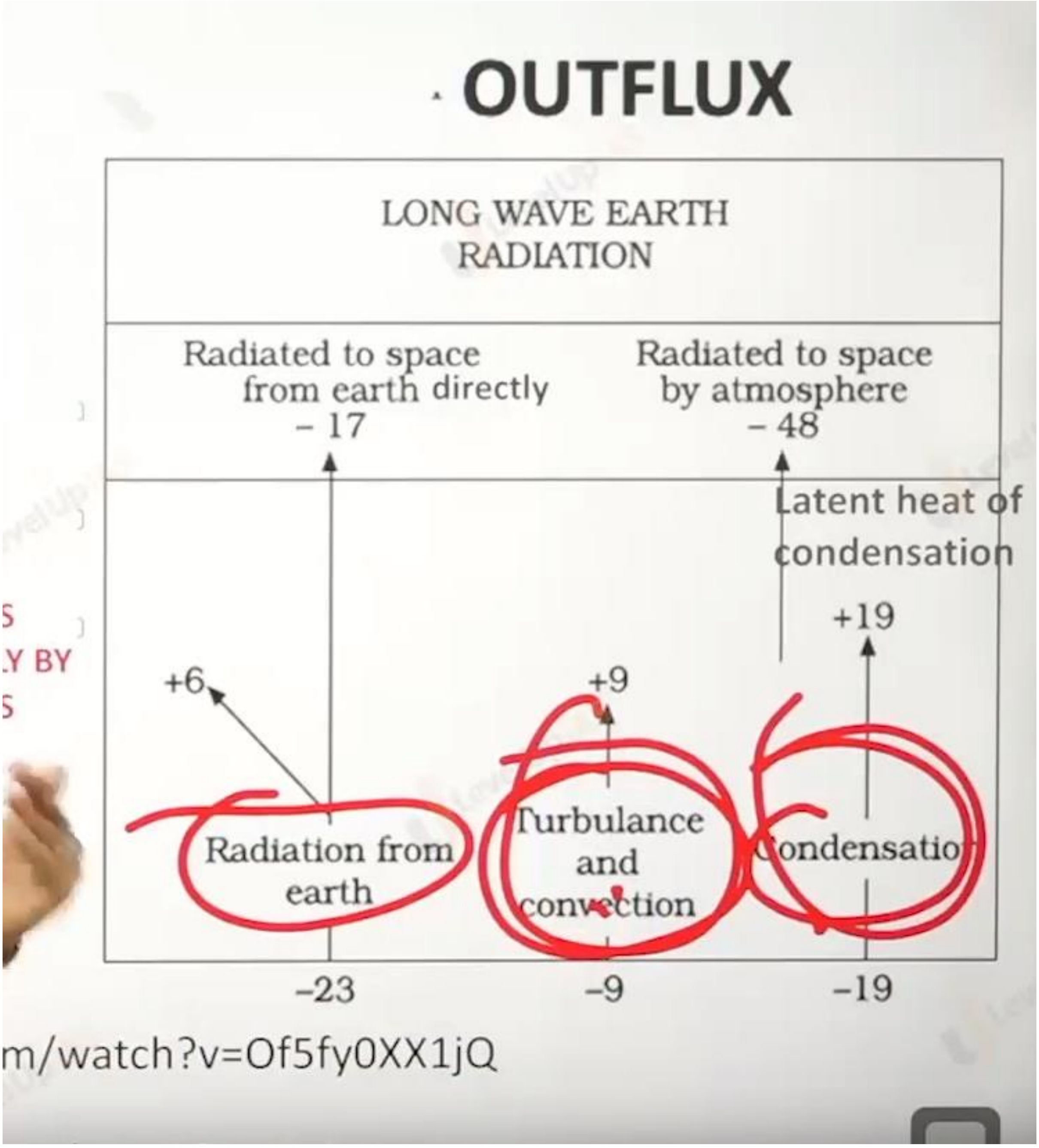
1. The earth's surface and the earth's atmosphere has variation in the temperature but on an average the earth's atmospheric temperature 15 to 17 degrees Celsius.
2. Earth's atmosphere has been capable of maintaining the average temperature i.e. earth is not getting progressively warmer and progressively cooler. So, the earth Is efficient heat engine. This maintenance of temperature can be defected by the heat budget.
3. Budget is a statement where it can be appreciated how energy comes and how the energy goes.
4. There are two components to heat budget
    a. Incoming solar radiation and outgoing radiation.




LevelUpIAS
HEAT BUDGET
Consider the insolation received at the top atmosphere is 100 units
INFLUX
OUTFLUX
SHORT WAVE SOLAR RADIATION
TOTAL SOLAR RADIATION
100 %
Scattered to space by Atm – 6
Reflected by space – 27 by clouds – 2 by earth
Scattered radiation
Direct raditaion
Clouds
+14 Absorbed by atmosphere
+17 Absorbed by the earth
+34 Absorbed by the earth
ABSORBED BY EARTH'S ATMOSPHERE DIRECTLY BY CONDUCTION: 6 UNITS
LONG WAVE EARTH RADIATION
Radiated to space from earth directly – 17
Radiated to space by atmosphere – 48
+6
+9
Radiation from earth
Turbulance and convection
–23
–9
https://www.youtube.com/watch?v=Of5fy0XX1jQ
ALBEDO IS ABOUT 29 UNITS
https://www.youtube.com/watch?v=Zt-fPFS3t-M

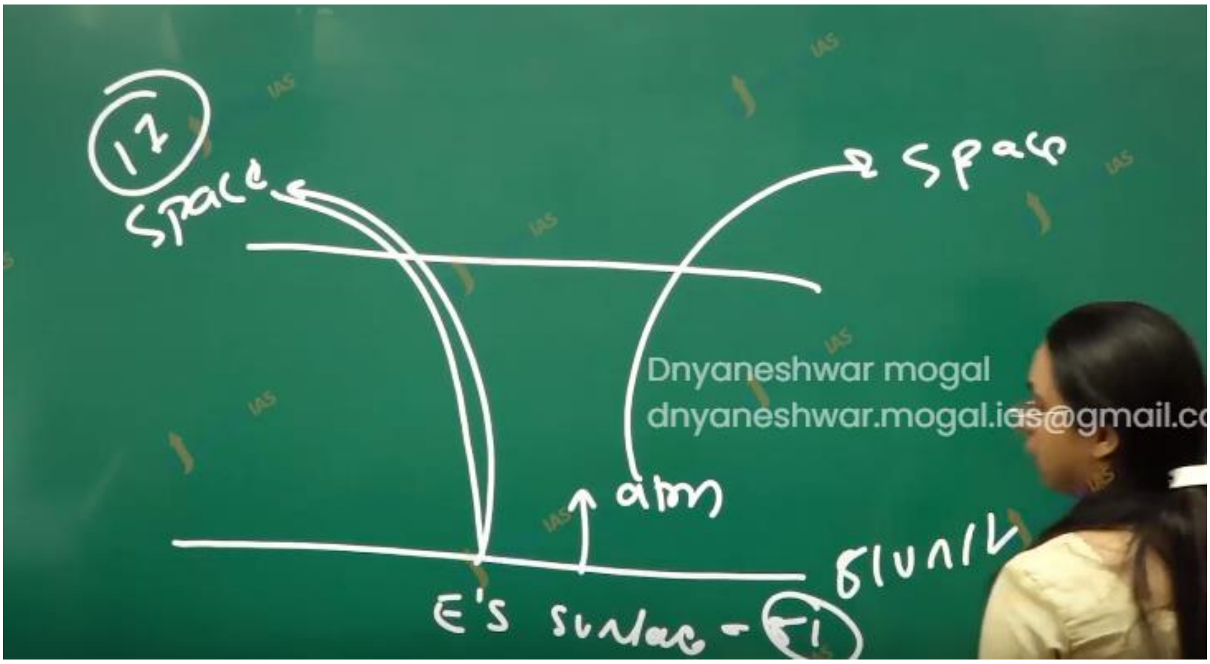

LevelUpIAS
INFLUX
SHORT WAVE SOLAR RADIATION
ALBEDO
space
TOTAL SOLAR RADIATION
100 %
Scattered to space by Atm – 6
Reflected by space – 27 by clouds – 2 by earth
15%-17%
Scattered radiation
Direct raditaion
Clouds
+14 Absorbed by atmosphere
+17 Absorbed by the earth
+34 Absorbed by the earth
ABSORBED BY EARTH'S ATMOSPHERE DIRECTLY B
CONDUCTION: 6 UNITS
https://www.youtube.com/


Temp pattern
Space
Atm
21
2
Dnyaneshwar mogal
dnyaneshwarmogal.ias@gmail.com



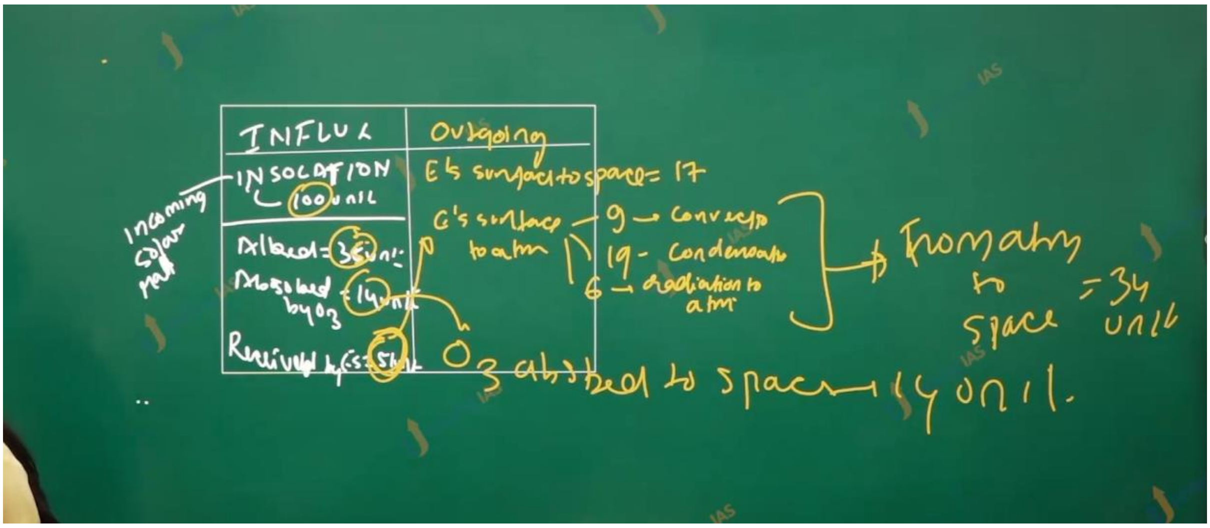
Influx component:

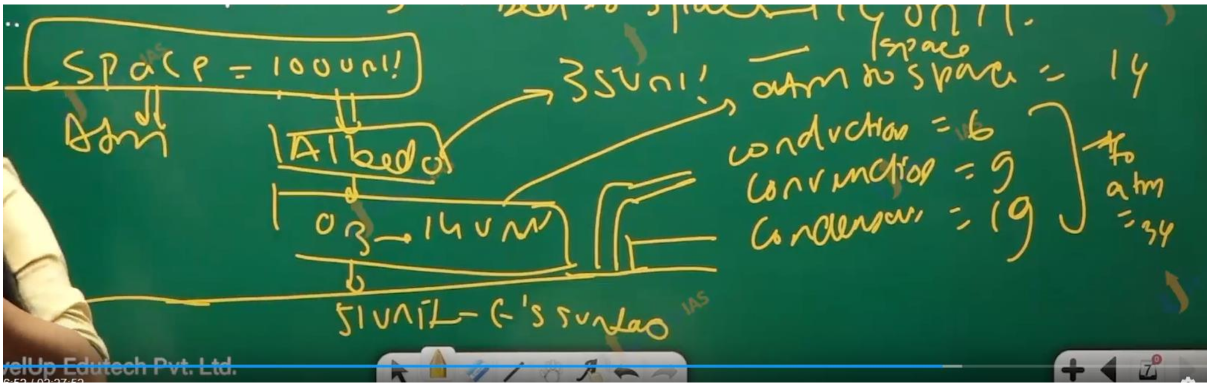
1. In flux 100 units are received from the solar radiation nearly 35% reflected (albedo reflectivity) 27% from the cloud 2 by the earth surface and 6 from the atmosphere.
2. 14 unit is observed by the ozone. And 51 units are observed by the earth's surface.



Outflux component.

1. 17 units are radiated from the earth’s surface directly to space.
2. The earth's surface transfer heat via radiation convection and condensation.
3. Surface to earth's atmosphere radiation losses is 6 units.
4. Turbulence and convection transfer are 9 units.
5. From earth’s surface to earth's atmosphere condensation related transfer are 19 unit
6. The units received by the atmosphere is 9+19+6 =34 Units.
7. Atmosphere sends 34 units to space. The 14 units observed by the ozone layer are sent to space by does completing the heat budget.



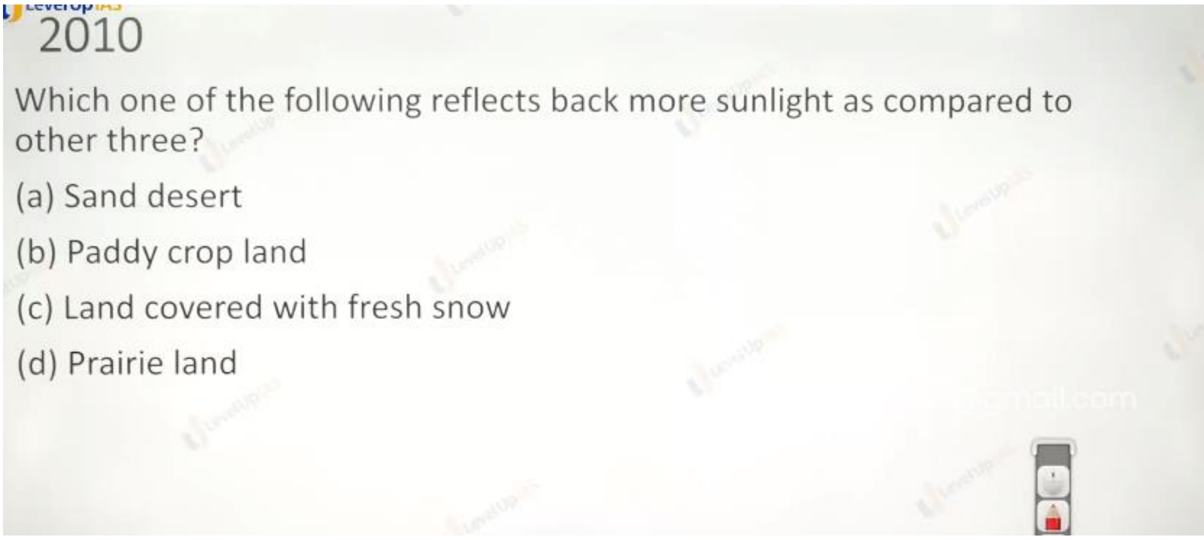


Heat budget :

1. heat receive is not lost instantaneously, the energy receive is observed and work is done, in formed of biological and other

physical processes this manifest in the form of climate and weather conditions.

2. Despite the earth atmosphere in the being balance there are places which are net receiver of energy, and there are places where energy is deficits.
3. The low latitude are latitudes of energy surplus while high latitude are energy deficits.
4. The internal latitudinal exchanges drives the atmosphere and ocean waterflow.



C

Answer: c

Sample Albedos of Earth

Surface	Typical albedo
Fresh asphalt	0.04
Open ocean	0.06
Worn asphalt	0.12
Conifer forest (Summer)	0.08, 0.09 to 0.15
Deciduous trees	0.15 to 0.18
Bare soil	0.17
Green grass	0.25
Desert sand	0.4
New concrete	0.55
Ocean ice	0.5–0.7
Fresh snow	0.80–0.90

2022

Q. **Consider the following statements :**

1. High clouds primarily reflect solar radiation and cool the surface of the Earth.
2. Low clouds have a high absorption of infrared radiation emanating from the Earth's surface and thus cause warming effect.

Which of the statements given above is/are correct ?

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

D. low cloud are thicker enough to reflect the more and it is white.so lower cloud reflect the more heat that high cloud.

And high cloud will observer more (don’t know.)

Role of cloud in warming and cooling the earth.

1. The low clouds primarily reflect the solar radiation and cool the surface of the earth.
2. High cloud observes the infrared the from the earth surface and thus have warming the planet.